

Renal & cardiovascular pharmacology

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Drugs acting on renal system

Function of renal system

- ✓ **Excretory function:** metabolic wastes, drugs, toxins, water soluble metabolites & hormones
- ✓ **Regulatory function:**
 - Water/fluid/ balance: diluting & concentrating urine
 - Electrolyte balance: K^+ , Na^+ , Ca^{2+} , Mg^{2+} , Cl^- , Pi^-
 - Acid base balance: H^+ , HCO_3^-
- ✓ **Endocrine function:** EPO, Renin, PGs
- ✓ **Metabolic function:** activation of Vit-D, gluconeogenesis

DIURETICS

- are drugs that increase the Volume of urine.

- **How Diuretics Work?**

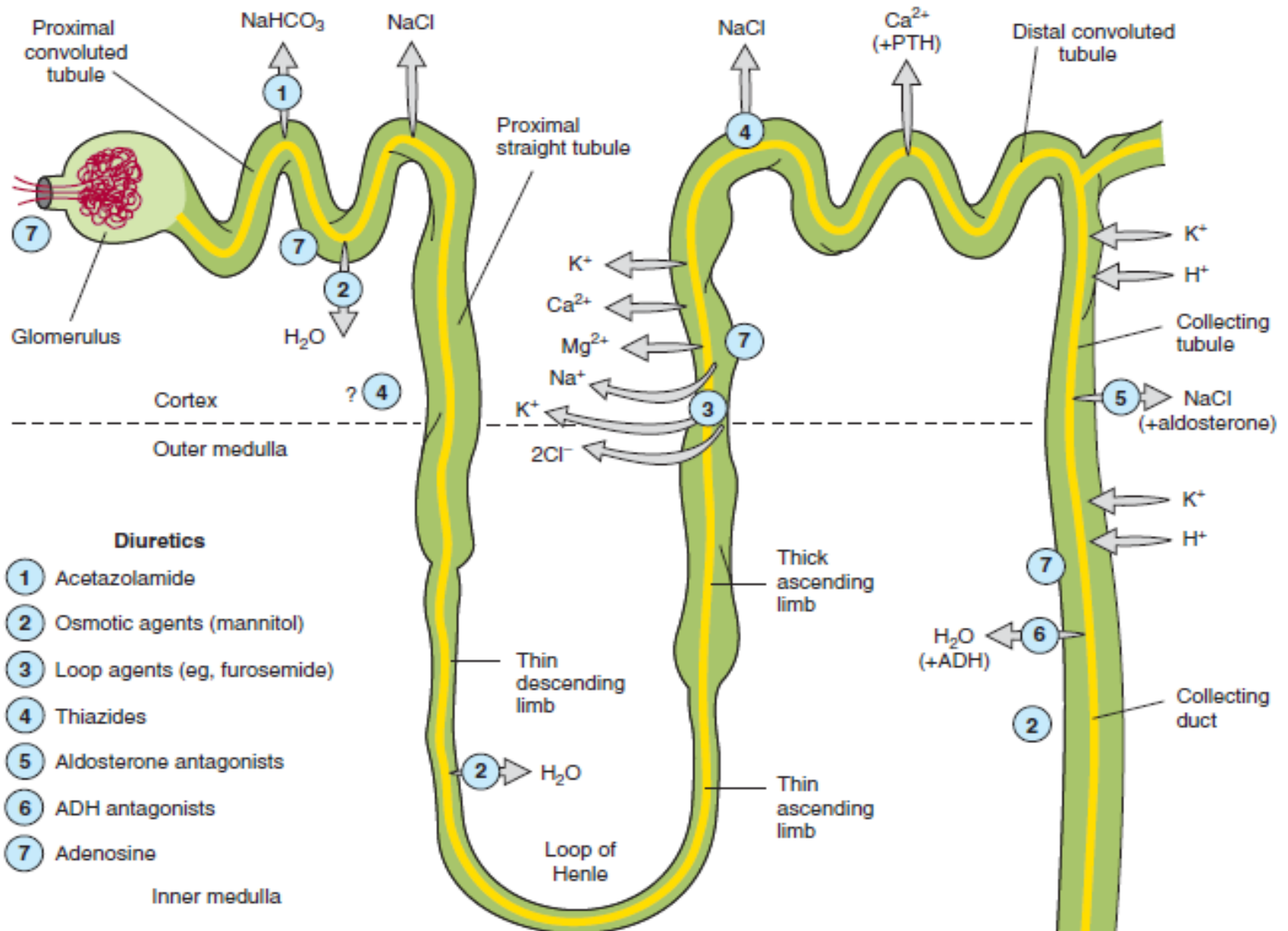
- ☛ blockade of sodium (and chloride) reabsorption

- mechanism of action for most diuretics

- ☛ Create an osmotic pressure within the nephron

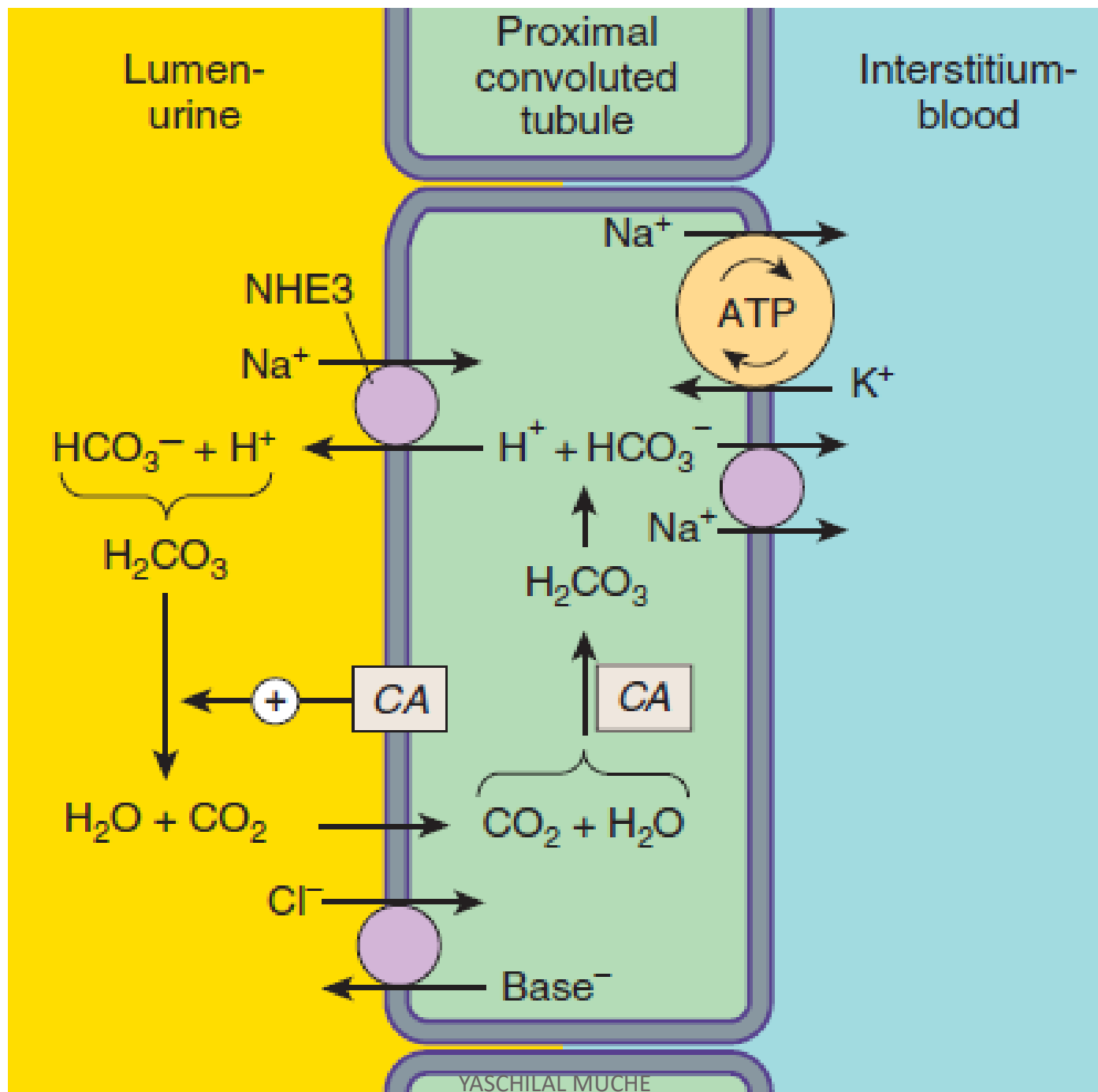
- » this prevents the passive reabsorption of water.

Segment	Functions	Water Permeability	Primary Transporters and Drug Targets at Apical Membrane	Diuretic with Major Action
Glomerulus	Formation of glomerular filtrate	Extremely high	None	None
Proximal convoluted tubule (PCT)	Reabsorption of 65% of filtered $\text{Na}^+/\text{K}^+/\text{CA}^{2+}$, and Mg^{2+} ; 85% of NaHCO_3 , and nearly 100% of glucose and amino acids. Isosmotic reabsorption of water.	Very high	Na/H^1 (NHE3), carbonic anhydrase; $\text{Na}/\text{glucose}$ cotransporter 2 (SGLT2)	Carbonic anhydrase inhibitors, Adenosine antagonists (under investigation)
Proximal tubule, straight segments	Secretion and reabsorption of organic acids and bases, including uric acid and most diuretics	Very high	Acid (eg, uric acid) and base transporters	None
Thin descending limb of Henle's loop	Passive reabsorption of water	High	Aquaporins	None
Thick ascending limb of Henle's loop (TAL)	Active reabsorption of 15–25% of filtered $\text{Na}^+/\text{K}^+/\text{Cl}^-$; secondary reabsorption of Ca^{2+} and Mg^{2+}	Very low	$\text{Na}/\text{K}/2\text{Cl}$ (NKCC2)	Loop diuretics
Distal convoluted tubule (DCT)	Active reabsorption of 4–8% of filtered Na^+ and Cl^- ; Ca^{2+} reabsorption under parathyroid hormone control	Very low	Na/Cl (NCC)	Thiazides
Cortical collecting tubule (CCT)	Na^+ reabsorption (2–5%) coupled to K^+ and H^+ secretion	Variable ²	Na channels (ENaC), K channels, ¹ H^+ transporter, ¹ aquaporins	K^+ -sparing diuretics Adenosine antagonists (under investigation)
Medullary collecting duct	Water reabsorption under vasopressin control	Variable ²	Aquaporins	Vasopressin antagonists ⁵



Carbonic Anhydrase Inhibitors

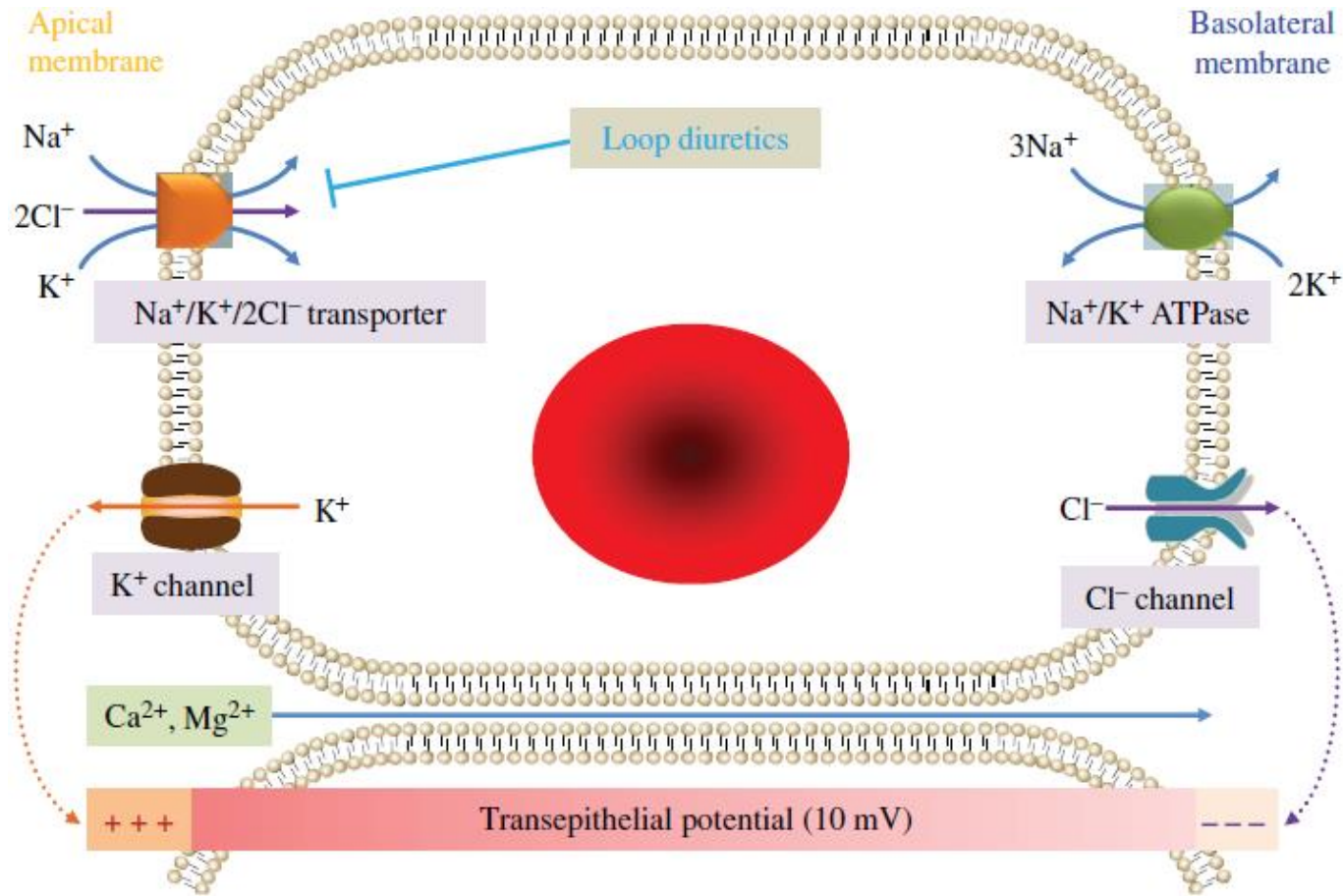
- **Acetazolamide , dorzolamide, Methazolamide, Dichlorphenamide**
 - MOA: reversible inhibitors of CA in kidney, eye, brain.
 - less effective diuretics than the others
 - Inhibition of carbonic anhydrase in the eye reduces intraocular pressure
- **Therapeutic uses;**
 - ✓ Treatment of glaucoma (dorzolamide & brinzolamide)
 - ✓ Mountain sickness (Acetazolamide)
 - ✓ Catamenial epilepsy (Acetazolamide)
 - ✓ Familial periodic paralysis (Dichlorphenamide)
- **Adverse effects;** Metabolic acidosis (mild), potassium depletion



High-Ceiling (Loop) Diuretics

- **Drugs:** furosemide, torsemide, Bumetanide, ethacrynic acid
 - Acts in thick segment of ascending limb of Henle's loop
 - inhibit $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ cotransporter
 - induce expression of the cyclooxygenase-2 (COX-2)
 - Have the greatest diuretic effect of all the diuretic drugs, since
 - the ascending limb accounts for reabsorption of 25% to 30% of filtered NaCl , and
 - downstream sites are unable to compensate for the increased Na^+ load.

Loop Diuretics...



- Mechanism: $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ transporter inhibition, results in: $\downarrow$ intracellular K^+ in TAL $\Rightarrow$ $\downarrow$ back diffusion of K^+ $\Rightarrow$ $\downarrow$ positive potential $\Rightarrow$ $\downarrow$ reabsorption of Ca^{2+} and Mg^{2+} (along with Na & K)

Loop Diuretics...

DRUG	RELATIVE POTENCY	ORAL AVAILABILITY	$t_{1/2}$ (hours)	ROUTE OF ELIMINATION
Furosemide	1	~60%	~1.5	~65% R, ~35% M ^a
Bumetanide	40	~80%	~0.8	~62% R, ~38% M
Ethacrynic acid	0.7	~100%	~1	~67% R, ~33% M
Torsemide	3	~80%	~3.5	~20% R, ~80% M
Azosemide ^b	1	~12%	~2.5	~27% R, ~63% M
Piretanide ^b	3	~80%	0.6–1.5	~50% R, ~50% M

M, metabolism; R, renal excretion of intact drug.

^aMetabolism of furosemide occurs predominantly in the kidney.

^bNot available in the U.S.

❑ Therapeutic uses of LD

- ✓ Acute pulmonary edema
- ✓ edema of hepatic, cardiac, or renal origin unresponsive to other drugs
- ✓ Hypertension

❑ Adverse effects

- ✓ Hyponatremia, hypochloremia, hypokalemia , dehydration, hypotension, ototoxicity, hyperglycemia, hyperuricemia (2° gout)
 - Potassium-sparing diuretics; counterbalance K wasting effects to decrease risk of hypokalemia

Thiazides and Thiazide-like Diuretics

- **Drugs:** chlorothiazide, hydrochlorothiazide, chlorthalidone, flumethazide, etc.
- includes diuretics that are **benzothiadiazine** derivatives (thiazide or thiazide-type diuretics) and drugs that are pharmacologically similar to thiazide diuretics but differ structurally (thiazide-like diuretics).
- All have equal efficacy except chlorothiazide (less effective),
 - They differ only in potency.

Thiazide & thiazide-like diuretics...

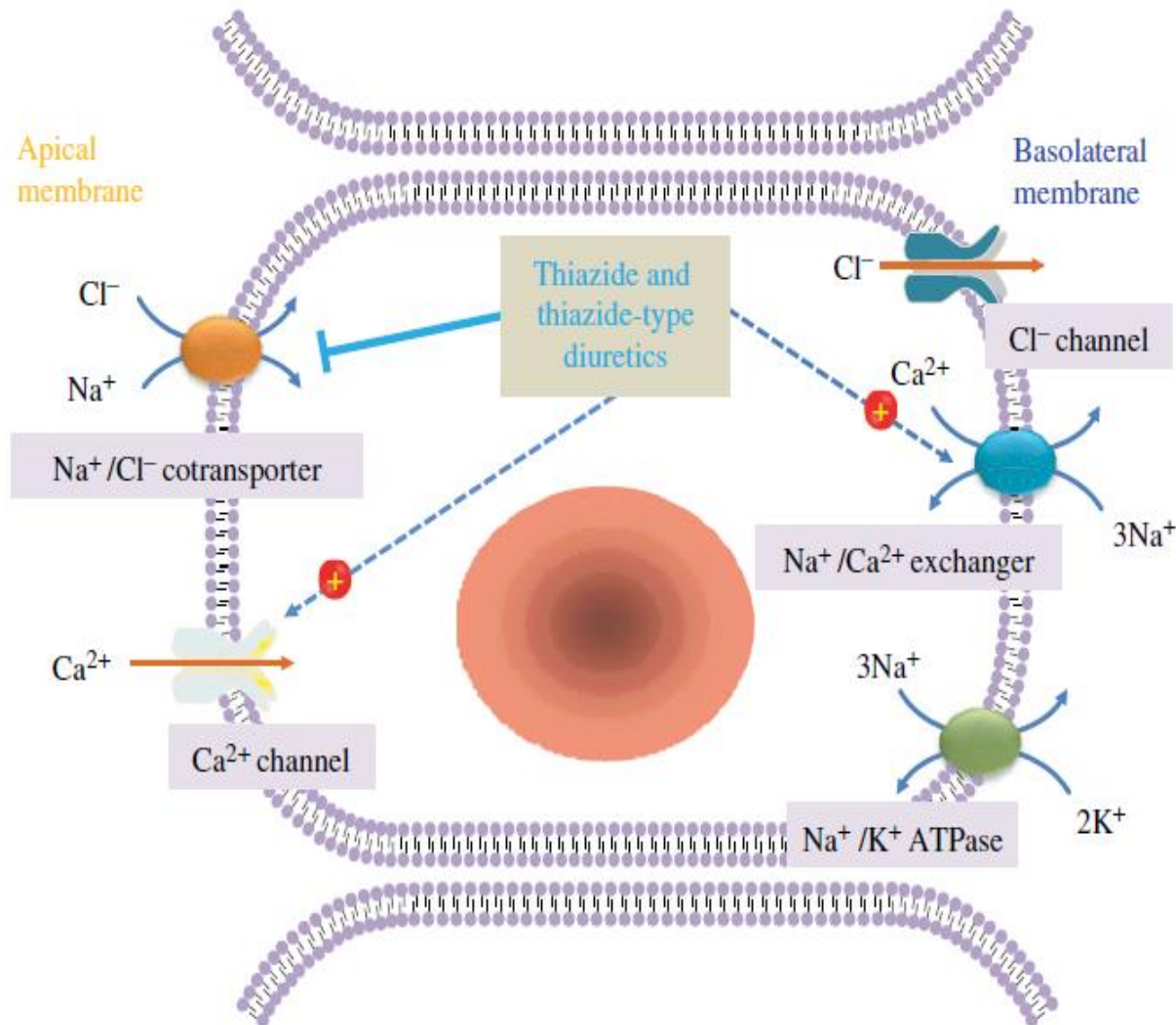
Thiazide diuretics

- ✓ Chlorothiazide
- ✓ Hydrochlorothiazide
- ✓ Bendroflumethiazide
- ✓ Methyclothiazide

Thiazide-like diuretics

- ✓ Chlorthalidone
- ✓ Indapamide
- ✓ Metolazone

Thiazide & thiazide-like diuretics...



Inhibit Na⁺/Cl⁻ co-transporter on the luminal membrane of DCT

Thiazide & thiazide-like diuretics...

Pharmacological effects

- are only moderately efficacious
 - $\uparrow$ Na^+ & Cl^- excretion (only 5% filtered Na^+ load)
- possess CA inhibition except indapamide
 - $\uparrow$ HCO_3^- & Phosphate excretion
- increase secretion of K^+ & H^+
- decrease Ca^{2+} excretion (chronic use)
- reduce uric acid excretion (long term use)
- loss of Mg^{2+} (long term use)

Therapeutic uses of Thiazide & thiazide-like diuretics

- Hypertension
- Edema associated with CHF and other conditions
 - Loop diuretics are preferred
 - Frequently combined with a loop diuretic to effect “sequential blockade” of tubular transport.
- Nephrolithiasis due to idiopathic hypercalciuria
 - Kidney stones caused by Ca^{2+} crystals
- Nephrogenic diabetes insipidus

ADR of Thiazide & thiazide-like diuretics

✓ **Electrolyte disturbance**

- Hypokalemia, Hyponatremia & Hypomagnesemia
- Hypercalcemia

✓ **Metabolic Abnormalities**

- Hyperlipidemias (except indapamide)
- Hyperglycemia (higher doses of HCT >50 mg/d, stimulate ATP-sensitive K⁺ channels of beta cells)
- Hyperuricemia (thiazides are the diuretics most associated with development of gout)

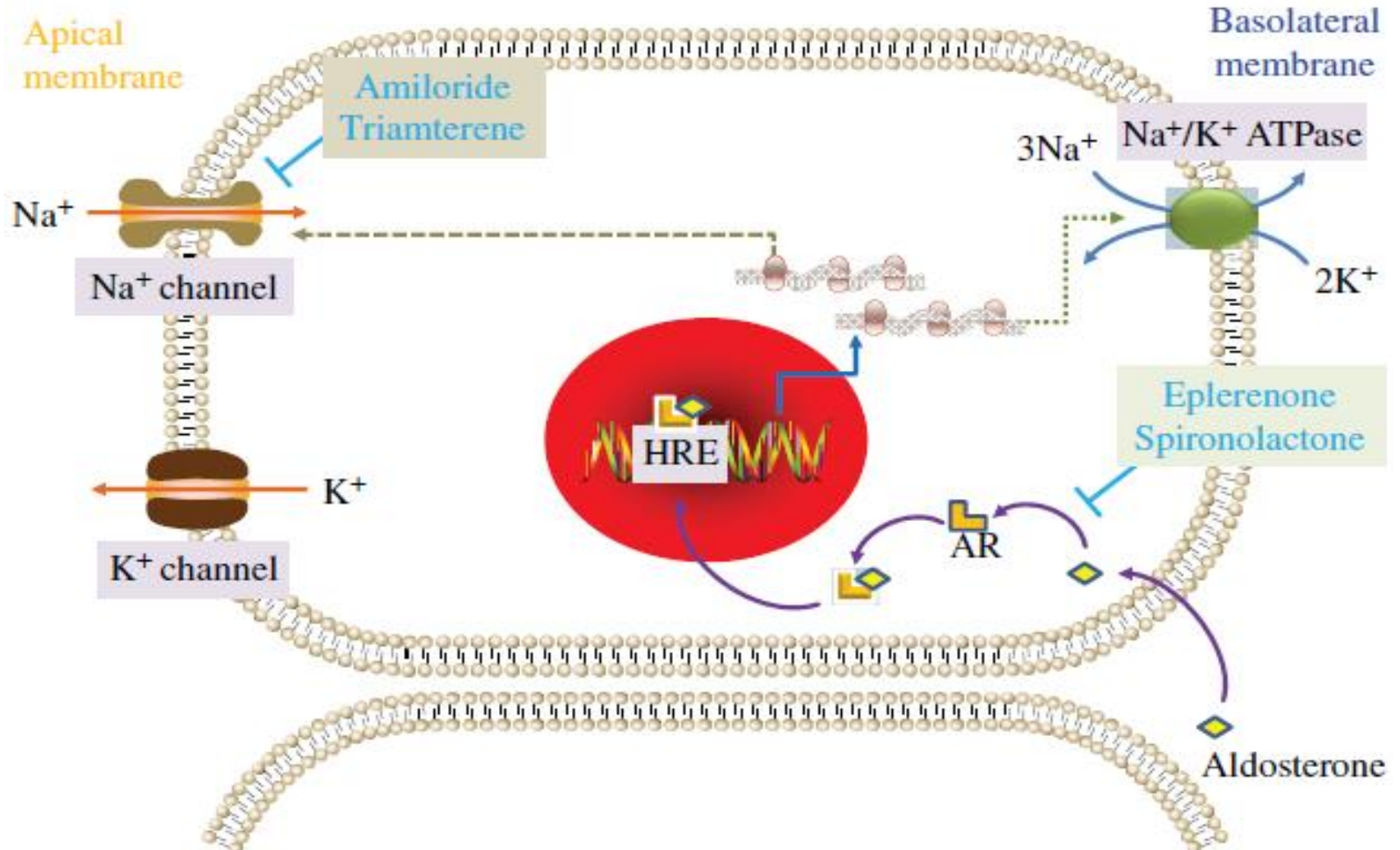
✓ **Allergic Reactions:** sulfonamide derivatives

✓ **Erectile dysfunction**

Potassium-Sparing Diuretics

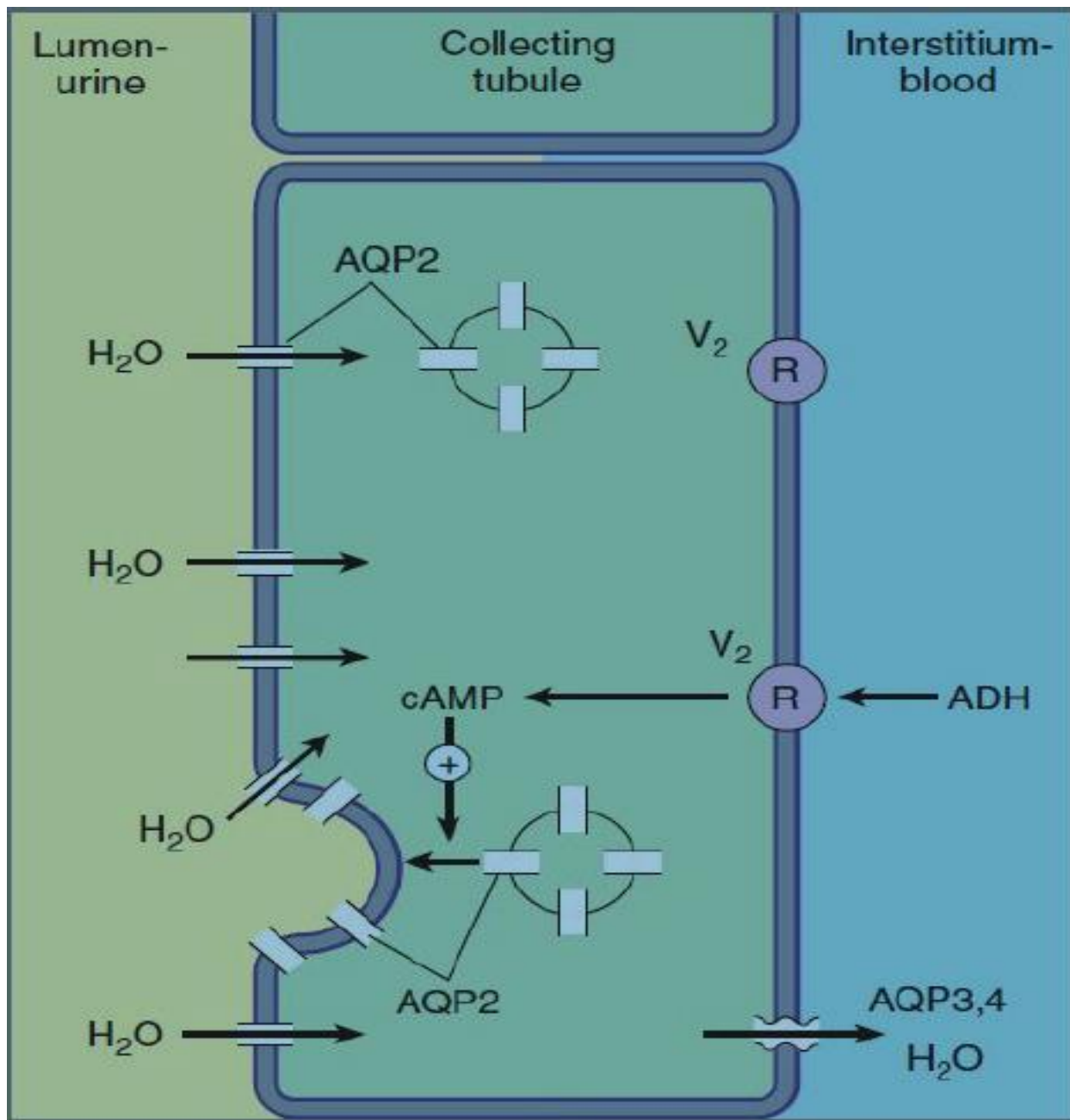
- Based on the mechanisms involved, classified into two subgroups:
 - (i) Direct Na^+ channel blockers (amiloride & triamterene)
 - (ii) Aldosterone receptor antagonists (eplerenone & spironolactone)
- **Therapeutic Uses**
 - For primary hyperaldosteronism
 - For hypertension and edema in combination with thiazide or loop diuretics
- **Adverse Effects;** Hyperkalemia, acidosis, antiandrogen effect.

K⁺ Sparing Diuretics...



ADH (Vasopressin) receptor antagonists

- There are multiple receptors for vasopressin (ADH)
 - the V1a, V1b, and V2 receptors.
- V1a and V1b receptors principally
 - cause vasoconstriction and mediate ACTH release, respectively
- The V2 receptors primarily mediate
 - the antidiuretic response
- The diuretics described in the preceding sections increase sodium and water excretion.
- In contrast, the vasopressin receptor antagonists (also called aquaretics)
 - inhibit the action of ADH, resulting in a selective water diuresis.



- These drugs are used for the treatment of hyponatremia,
 - since water loss will raise the serum sodium concentration.
- ***Tolvaptan , mozavaptan, satavaptan, and lixivaptan***
 - selective for the V2 receptor
- ***conivaptan***
 - blocks both the V2 and V1a receptors

Osmotic Diuretics

- Water soluble and are hence freely filtered
- Insoluble in lipid and hence are poorly reabsorbed
- Site of action: Nephron segments which are freely permeable to water

❑ **Drugs:** Mannitol, Urea, Glycerin, Isosorbide

- Glycerin & Isosorbide: orally effective
- Mannitol & urea: orally ineffective (Given IV)

➡ **Therapeutic uses**

- » reduction of intracranial pressure (ICP), and
- » reduction of intraocular pressure

SUMMARY

Group	Urinary Electrolytes			Body pH
	NaCl	NaHCO ₃	K ⁺	
Carbonic anhydrase inhibitors	+	+++	+	↓
Loop agents	++++	0	+	↑
Thiazides	++	+	+	↑
Loop agents plus thiazides	+++++	+	++	↑
K ⁺ -sparing agents	+	(+)	–	↓

+, increase; –, decrease; 0, no change; ↓, acidosis; ↑, alkalosis.

Class	Adverse Side Effects	Drug Interactions
Thiazide	Hypokalemia , hyponatremia, metabolic alkalosis, dehydration (hypovolemia), hypotension, hyperglycemia in diabetics, hyperuricemia, hypercholesterolemia; hypertriglyceridemia, Increased LDL	hypokalemia ↑ digitalis toxicity NSAIDs: ↓ diuretic efficacy corticosteroids: ↑ hypokalemia
Loop	hypokalemia , hypo-magnesemia, metabolic alkalosis, hyperuricemia, hypo-volemia, hypotension, dose-related hearing loss (ototoxicity)	hypokalemia ↑ digitalis toxicity NSAIDs: ↓ diuretic efficacy corticosteroids: ↑ hypokalemia aminoglycosides: ↑ ototoxicity, nephrotoxicity
K ⁺ -sparing	Hyperkalemia , metabolic acidosis, gynecomastia , gastric problems including peptic ulcer	ACE inhibitors: ↑ hyperkalemia NSAIDs: ↓ diuretic efficacy
CA inhibitors	Hypokalemia , metabolic acidosis	

Case study

- A 65-year-old man has a history of diabetes and chronic kidney disease with baseline creatinine of 2.2 mg/dL. Despite five different antihypertensive drugs, his clinic blood pressure is 176/92 mm Hg; he has mild dyspnea on exertion and 2–3+ edema on exam. He has been taking furosemide 80 mg twice a day for 1 year now. At the clinic visit, hydrochlorothiazide 25 mg daily is added for better blood pressure control and also to treat symptoms and signs of fluid overload. Two weeks later, the patient presents to the emergency department with symptoms of weakness, anorexia, and generalized malaise. His blood pressure is now 91/58 mm Hg, and he has lost 15 kg in 2 weeks. His laboratory tests are significant for a serum creatinine of 10.8 mg/dL.
- What has led to the acute kidney injury?
- What is the reason for the weight loss?
- What precautions could have been taken to avoid this hospitalization?

Inhibitors of the renin- angiotensin-Aldosterone system

RAAS

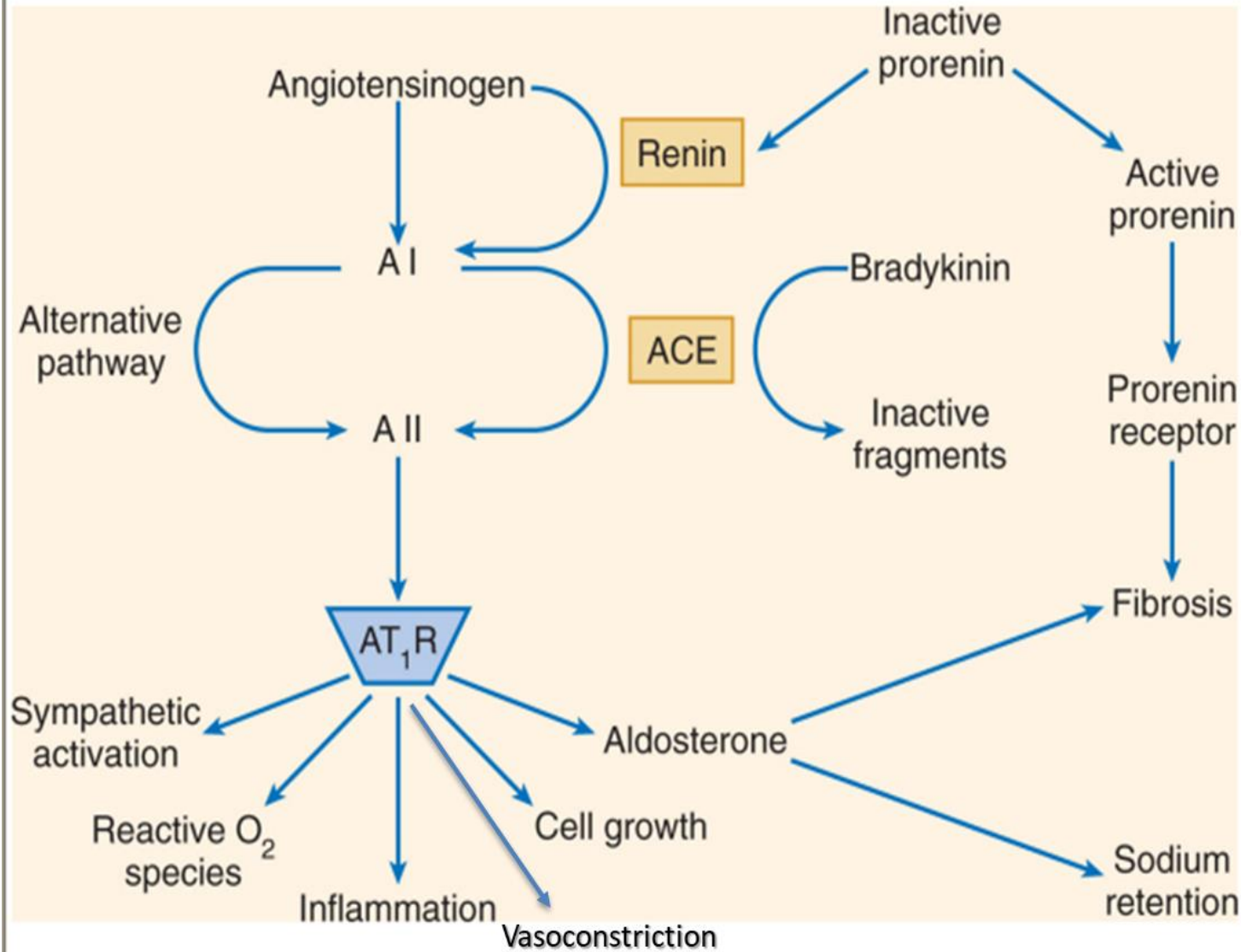
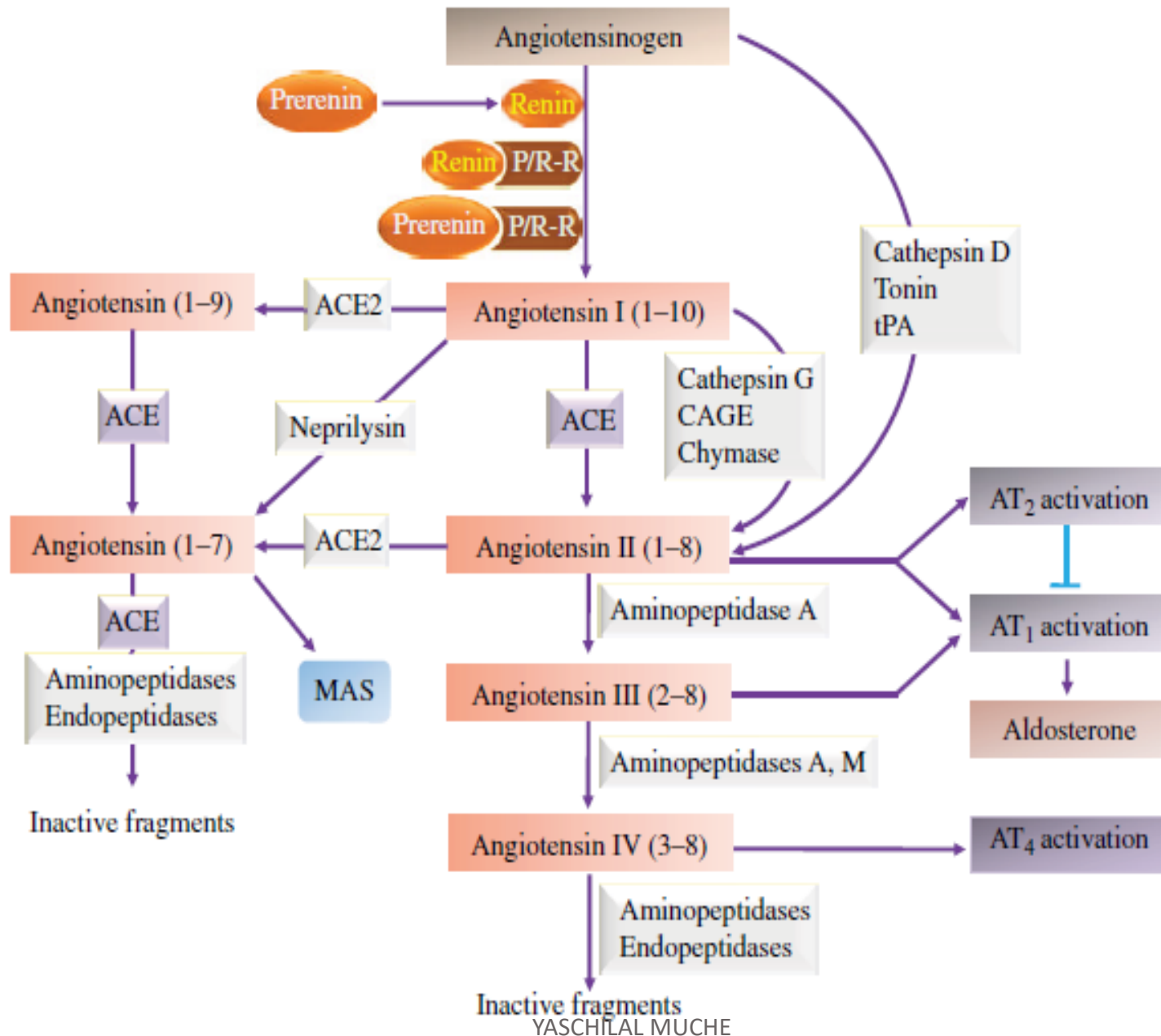


FIGURE 40-9 The renin-angiotensin-aldosterone system. A I = angiotensin I; A II = angiotensin II; ACE = angiotensin-converting enzyme; AT₁R = angiotensin I receptor.

Alternative Pathways for Angiotensin II Biosynthesis

- Angiotensin II may be generated through ACE-independent pathways or “ACE escape.”
 - Angiotensinogen is directly converted to AngII by
 - ✓ **cathepsin D** and **tonin**.
 - Enzymes that convert AngI to AngII include
 - ✓ **cathepsin G, chymostatin-sensitive AngII-generating enzyme, and chymase.**

The contemporary view of the RAAS

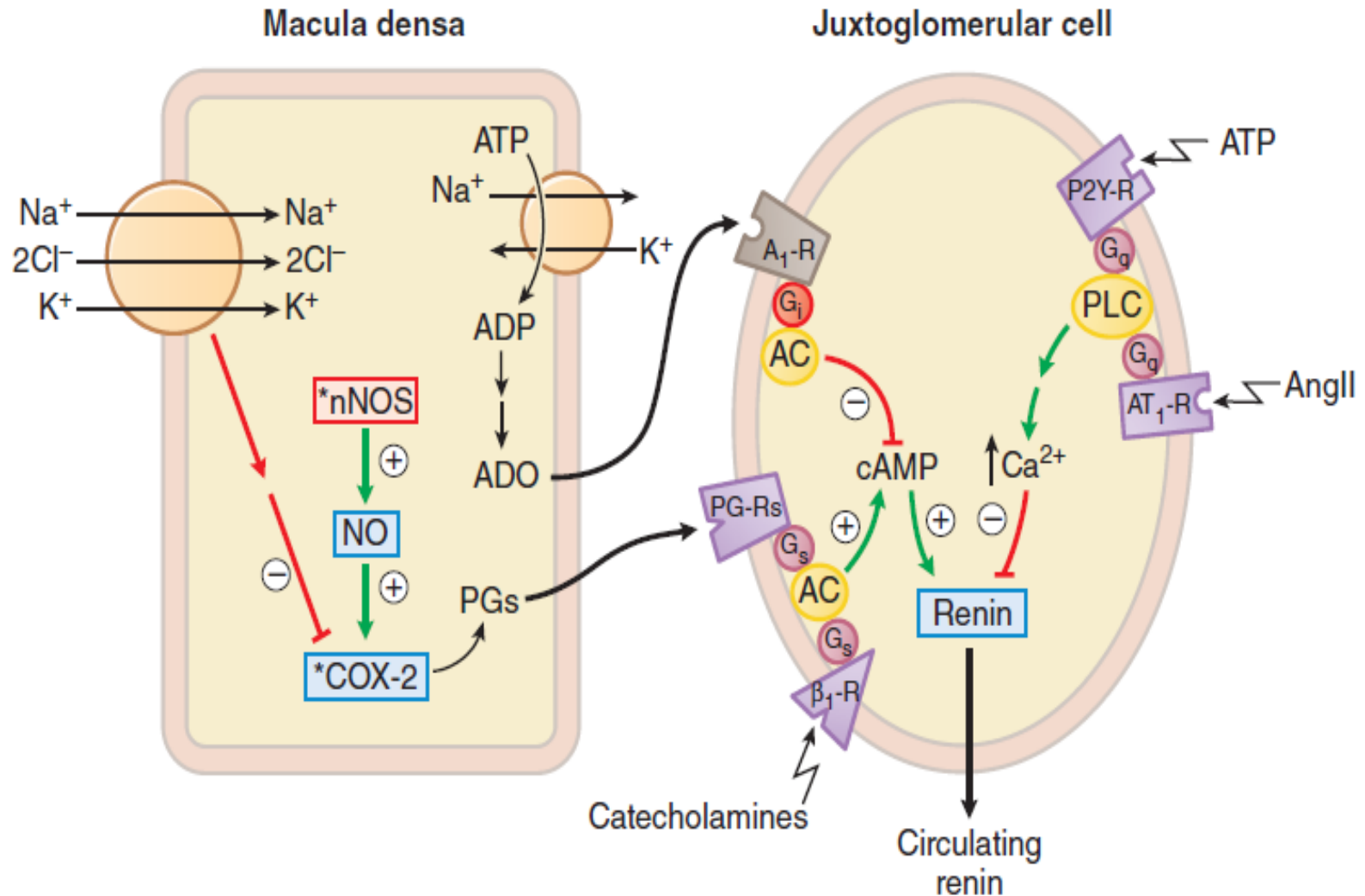


Control of Renin Secretion:

- Renin is secreted by the granular cells within the JG apparatus and is regulated by the following pathways:
 - I. The macula densa pathway
 - II. The intrarenal baroreceptor pathway
 - III. The β_1 adrenergic receptor pathway

I. Regulation of JG cell renin release by the macula densa

ADO=adenosine



Control of Renin Secretion...

II. The Intrarenal baroreceptor pathway

- Increased in renal perfusion pressure in the preglomerular vessels inhibit renin release and vice versa.

III. The β_1 adrenergic receptor pathway

- Activation of β_1 receptors on JG cells increases cyclic AMP and enhances renin secretion.
- Treatment with β blockers reduces renin secretion and PRA.

Feedback Regulation of renin:

- ✓ Short-loop negative feedback

- renin secretion enhances the formation of AngII, which stimulates AT1 receptors on JG cells to inhibit renin release.

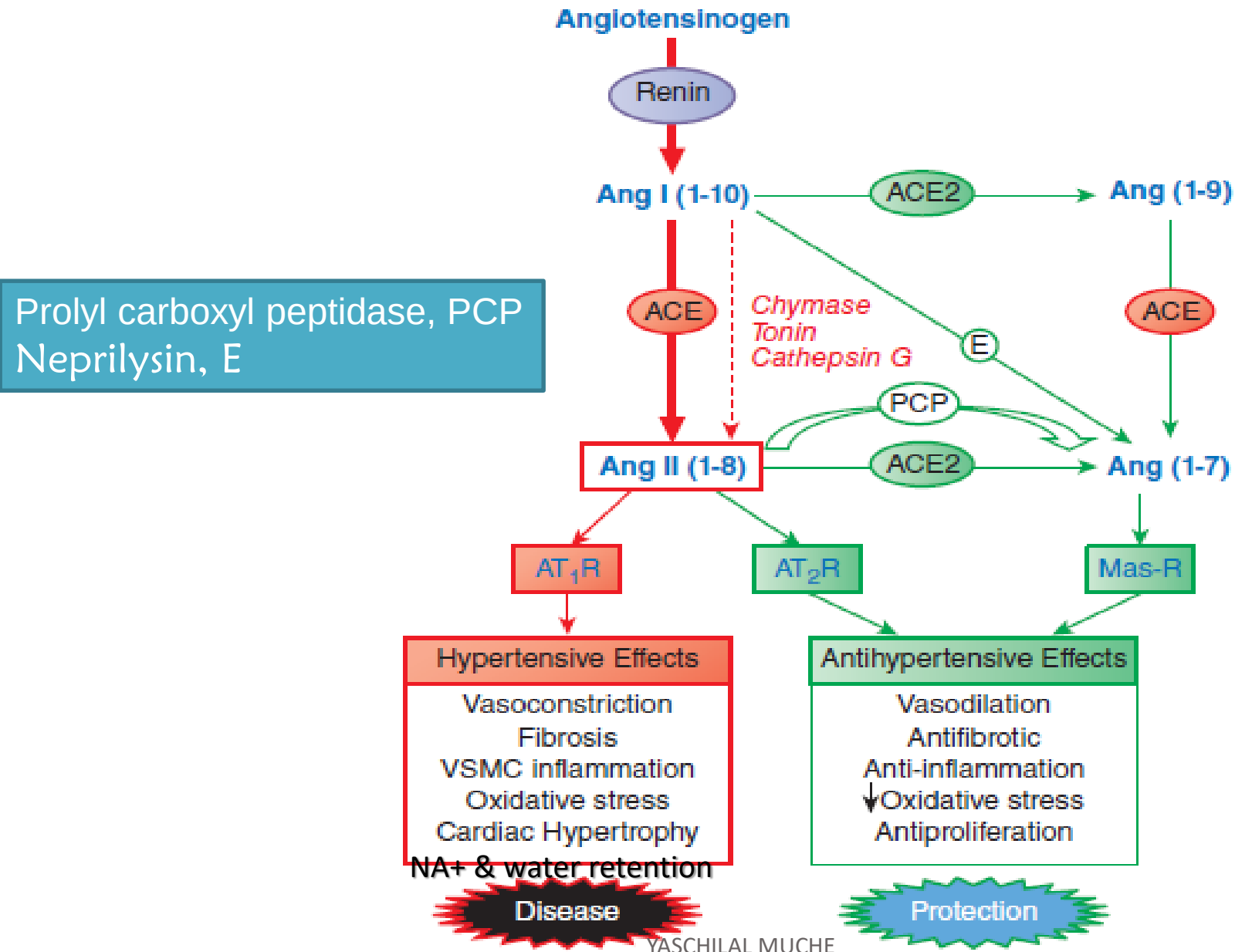
- ✓ long-loop negative feedback

- Inhibition of renin release due to AngII-dependent increase in blood pressure.

ACE₂

- Convert AngII to Ang(1–7) & also convert Ang I to Ang(1–9)
- AngII is the preferred substrate for ACE₂,
 - with 400-fold higher affinity than AngI
- Not inhibited by the standard ACEIs
- The counter-regulation of the actions of AngII by ACE₂ occur in at least two ways:
 - It decreases AngII levels metabolizing it to Ang(1–7).
 - It increases levels of Ang(1–7), which acts on Mas receptors to oppose AngII actions

Opposing arms in the RAS



Angiotensin II

Altered
Peripheral
Resistance

Altered
Renal
Function

Altered
Cardiovascular
Structure

MECHANISMS

- I. Direct vasoconstriction
- II. Enhancement of peripheral noradrenergic neurotransmission:
 - A. Increased NE release
 - B. Decreased NE reuptake
 - C. Increased vascular responsiveness
- III. Increased sympathetic discharge (CNS)
- IV. Release of catecholamines from adrenal medulla

- I. Direct effect to increase Na^+ reabsorption in proximal tubule
- II. Release of aldosterone from adrenal cortex (increased Na^+ reabsorption and increased K^+ excretion in distal nephron)
- III. Altered renal hemodynamics:
 - A. Direct renal vasoconstriction
 - B. Enhanced noradrenergic neurotransmission in kidney
 - C. Increased renal sympathetic tone (CNS)

- I. Nonhemodynamically mediated effects:
 - A. Increased expression of proto-oncogenes
 - B. Increased production of growth factors
 - C. Increased synthesis of extracellular matrix proteins
- II. Hemodynamically mediated effects:
 - A. Increased afterload (cardiac)
 - B. Increased wall tension (vascular)

RESULTS

Rapid Pressor Response

Slow Pressor Response

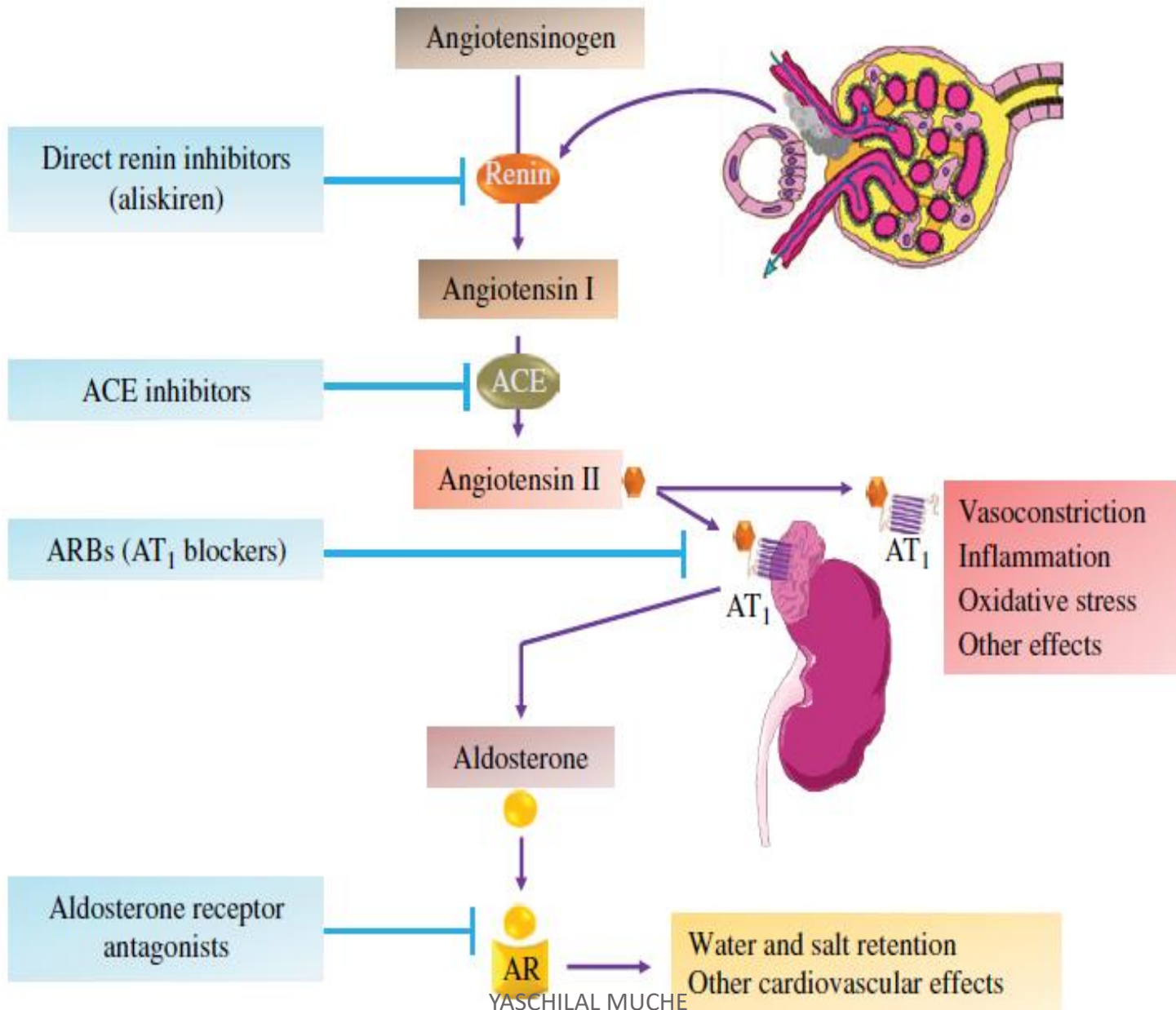
YASCHILAL MUCHE

Vascular and Cardiac Hypertrophy and Remodeling

Classes of drugs targeting RAAS

1. Direct renin inhibitors (DRI)
2. Angiotensin-converting enzyme (ACE) inhibitors
3. Angiotensin receptor blockers (ARBs)
4. Aldosterone receptor antagonists

Targeting sites of the RAAS inhibitors:



Direct Renin Inhibitors

Drugs: Aliskiren, remikiren and enalkiren

❖ Aliskiren

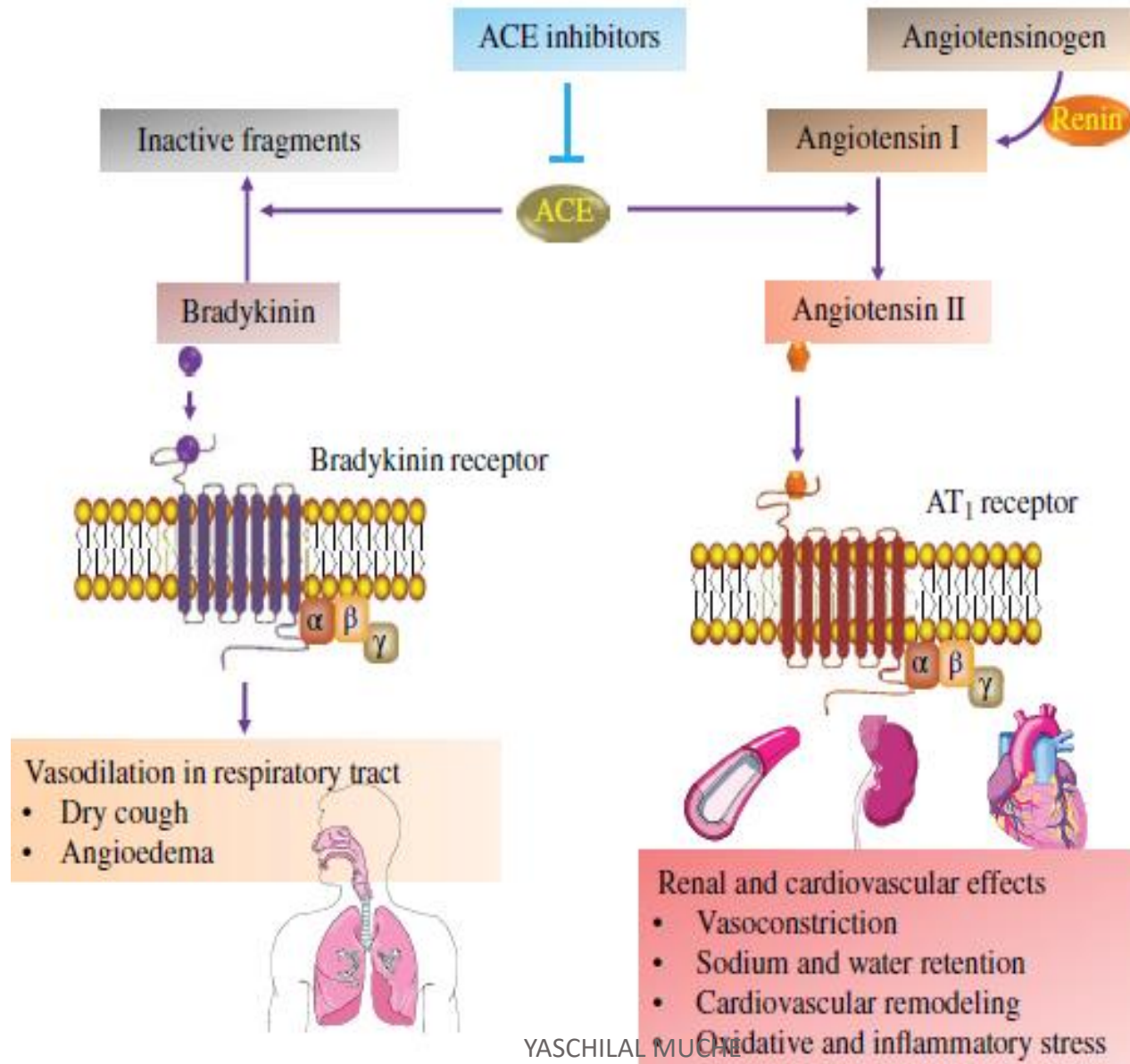
- is the only DRI approved for clinical use.
- reduces AngI, AngII & aldosterone levels
- a substrate for P-glycoprotein (low bioavailability, ~2.5%).
- $t_{1/2}$ is 20–45 h; Has high affinity and potency
- fatty meals significantly decrease its absorption
- reduces absorption of furosemide by 50%
- Irbesartan reduces the C_{max} of aliskiren by 50%.
- Elimination is mostly as unchanged drug in feces

Clinical use of Aliskiren

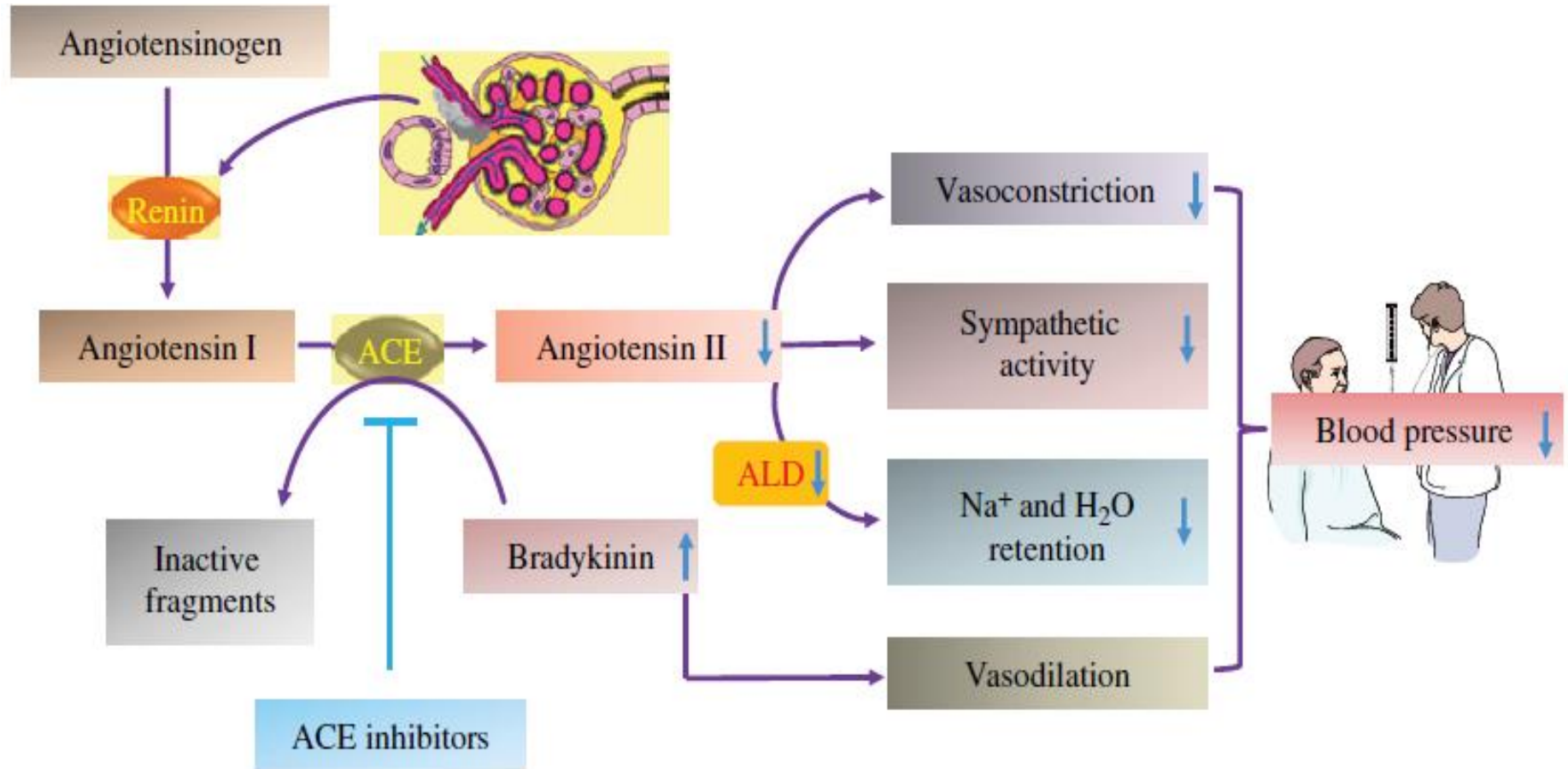
✓ Hypertension

- Therapeutic value unclear; no evidence for superiority over ACEIs or ARBs
- FDC (Valsartan+ Aliskiren) for hypertension not adequately controlled by other medications.
- FDC: Aliskiren + HCT; Aliskiren + Amlodipine
- Common ADR: mild diarrhea and GI problems
- Rare, serious adverse effects: angioedema and renal dysfunction
- Contraindicated in pregnancy or renal insufficiency
- Combination with RAS inhibitors is contraindicated

ACE inhibitors



ACE inhibitors...



ACE inhibitors...

Pharmacological Effects

- inhibit AngII production (decrease all Ag II mediated effects)
- will increase renin release and the rate of formation of AngI
 - by interfering with both short and long-loop negative feedbacks on renin release.
- accumulating AngI is directed down alternative metabolic routes, resulting in
 - the increased production of vasodilator peptides such as Ang(1–9) and Ang(1–7).

Classification

- **Three broad groups based on chemical structure:**
 - I. Sulfhydryl-containing ACE inhibitor: Captopril
 - II. Phosphorus-containing ACE inhibitor: Fosinopril
 - III. Dicarboxyl-containing ACE inhibitors (others)
- Many ACE inhibitors are ester-containing prodrugs that are 100-1000 times less potent but have better oral bioavailability than the active molecules.

ACE inhibitors...

Predominantly eliminated by the kidneys (need dose adjustment in renal failure)

- ✓ Captopril
- ✓ Lisinopril
- ✓ Enalapril
- ✓ Benazepril
- ✓ Quinapril
- ✓ Ramipril
- ✓ Moexipril
- ✓ Enalaprilat (IV)

Undergo both hepatic and renal elimination.
Not significantly altered by renal impairment.

- ✓ Fosinopril
- ✓ Trandolapril
- ✓ Perindopril

Except for captopril (with a duration of action of 6–12h), Most ACE inhibitors have a duration of action of ~24h, enabling the once daily dosing.

ACE inhibitors...

- All ACEI are prodrugs except captopril and lisinopril.
- Other drugs like enalapril are converted to its active metabolite (enalaprilat) and thus are slow acting.
- Enalaprilat is available for use in hypertensive emergencies by i.v. route.
- Important differences between captopril and other ACEIs is that
 - captopril is less potent, has fast onset and short duration of action and less absorption in presence of food in GIT.
 - Because of short and fast action, it can cause postural hypotension which is not seen with other ACEIs.

Therapeutic Uses of ACE Inhibitors

- ACEIs are used for the treatment of
 - ✓ Hypertension
 - ✓ CHF
 - ✓ MI
 - ✓ diabetic nephropathy
 - ✓ Chronic renal disease
 - ✓ scleroderma crisis
- These drugs reduce proteinuria in diabetic as well as non-diabetic chronic renal disease and also prevent the manifestations of scleroderma crisis which are mediated by angiotensin II.

Therapeutic Uses...

✓ Diabetic nephropathy and chronic Renal Failure

- Two mechanisms underlying ACE inhibitor-mediated renal protection

① **Pressure mechanism:**

- Increased glomerular capillary pressure is an important factor leading to glomerular injury.
- ACE inhibitors reduce glomerular capillary pressure by decreasing the arterial blood pressure and by dilating the renal efferent arterioles.

Therapeutic Uses...

② Inflammation and remodeling mechanism:

- Angiotensin II causes inflammation, oxidative stress, mesangial cell proliferation, and renal tissue remodeling, all of which contribute to the development of diabetic nephropathy.
- ACE inhibitors attenuate the aforementioned deleterious processes by decreasing the formation of angiotensin II.

Adverse Effects of ACE Inhibitors

- ✓ **Hypotension**, following the first dose
- ✓ **Dry cough** (5%–20% of patients)
 - mediated by the accumulation in the lungs of bradykinin, substance P, or PGs.
 - Thromboxane antagonism, aspirin, and iron supplementation reduce cough induced by ACE inhibitors.
 - ACE dose reduction or switching to an ARB is sometimes effective.
 - Once ACE inhibitors are stopped, the cough disappears, usually within 4 days.

Adverse Effects of ACE Inhibitors...

✓ Hyperkalemia

- Significant K^+ retention is rarely encountered in patients with normal renal function.
- However, ACE inhibitors may cause hyperkalemia
 - in patients with renal insufficiency or diabetes
 - in patients taking K^+ -sparing diuretics, K^+ supplements, β receptor blockers, or NSAIDs.

Adverse Effects of ACE Inhibitors...

✓ Acute Renal Failure

- ACEIs can induce acute renal insufficiency in patients with
 - bilateral renal artery stenosis, stenosis of the artery to a single remaining kidney, heart failure, or volume depletion owing to diarrhea or diuretics.
- An increase in serum creatinine upto 30% within 2-5 days can be expected in most patients started on ACEIs.
- It stabilizes in 2-3 weeks and is reversible on stopping
- Contraindicated when serum creatinine is >3.5 mg/dl.

Adverse Effects of ACE Inhibitors...

- ✓ **Angioedema** (0.1%–0.5% of patients), rare but fatal
 - Due to increased bradykinin
 - Rapid swelling in the nose, throat, mouth, glottis, larynx, lips, or tongue.
 - disappears within hours up on drug withdrawal
 - African Americans have a 4.5 times greater risk than Caucasians.
- ✓ **Fetopathic Potential**, may be due in part to fetal hypotension

Drug Interactions

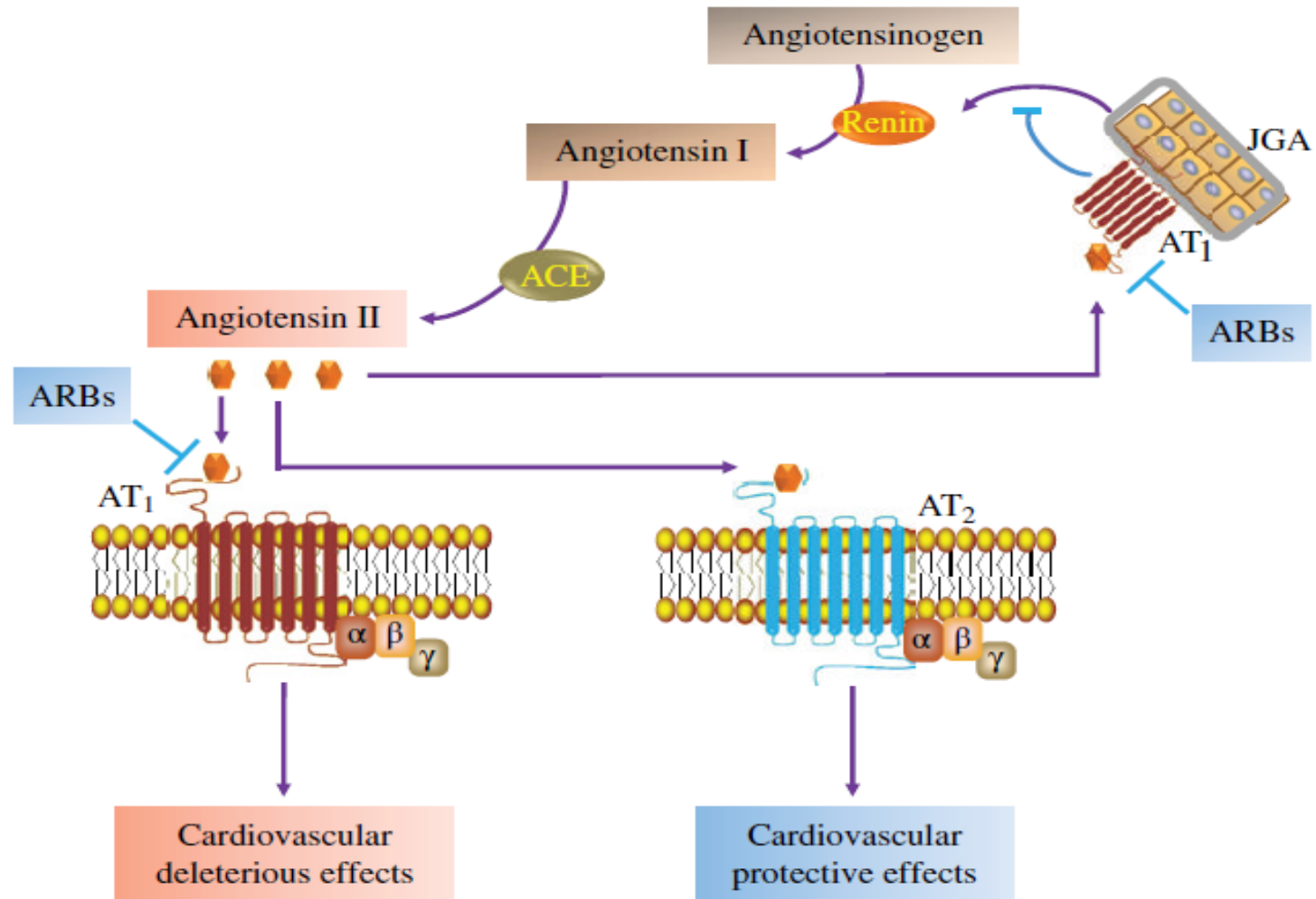
- Antacids may reduce the bioavailability of ACE inhibitors
- NSAIDs, including aspirin, may reduce the antihypertensive response to ACE inhibitors
- K^+ -sparing diuretics and K^+ supplements
 - exacerbate ACE inhibitor–induced hyperkalemia
- ACE inhibitors may increase plasma levels of digoxin

Angiotensin II Receptor Blockers

- ✓ Losartan
- ✓ Valsartan
- ✓ Eprosartan
- ✓ Irbesartan
- ✓ Candesartan
- ✓ Olmesartan
- ✓ Telmisartan
- ✓ Azilsartan

- Oral bioavailability of ARBs generally is low (<50%) except for azilsartan (~60%) and irbesartan (~70%), and protein binding is high (>90%).
- Vary widely in their volume of distribution from 10l for candesartan to 500l for telmisartan
- Most of the drugs are dosed once daily

Mechanisms of action of ARBs



ARBs VS ACEIs

- ARBs reduce activation of AT_1 receptors more effectively than do ACE inhibitors
 - ✓ ARBs block the actions of AngII via the AT_1 receptor regardless of the biochemical pathway leading to AngII formation.
- In contrast to ACE inhibitors, ARBs permit activation of AT_2 receptors.
 - ✓ Because ARBs block AT_1 receptors, increased level of AngII is available to activate AT_2 receptors.
- ACE inhibitors and ARBs increase effects of Ang(1–7) by different mechanisms.
- ACE inhibitors increase the levels of a number of ACE substrates, including bradykinin.

Pharmacological Effects:

- ARBs bind to the AT₁ receptors with high affinity (10,000 xAT₂)
- Inhibit Angiotensin II-induced deleterious effects
- No effects on bradykinin and reduced incidence of AEs
- Blockade of AT₁ with ARBs
 - Increased activation of the AT₂ receptors, potentially leading to additional cardiovascular protection.
- Telmisartan, but not the other ARBs, reduces triglyceride levels and improves insulin sensitivity possibly via
 - activation of peroxisome proliferator- activated receptor-gamma (PPAR γ)
- Both telmisartan and losartan have antiplatelet aggregation activity.

Therapeutic uses

- The clinical uses of ARBs include the treatment of the following diseases:
 - ✓ Hypertension (efficacy comparable to ACEIs with better safety)
 - ✓ Chronic Heart failure (Valsartan and Candesartan)
 - ✓ Diabetic nephropathy (Irbesartan and losartan)
 - ✓ Chronic renal failure
 - ✓ Stroke prophylaxis (losartan, but not effective in black race)

Adverse Effects

- ✓ The ARBs are generally well tolerated.
- ✓ The incidence of angioedema and cough with ARBs
 - is less than that with ACE inhibitors.
- ✓ Adverse effects include hyperkalemia and hypotension
- ✓ Contraindicated in patients with renal insufficiency
- ✓ Should be stopped during pregnancy

Drugs for Hypertension

Hypertension

- Hypertension is defined as persistently elevated arterial blood pressure $\geq 130/80$ mm Hg.

- If not detected early and treated appropriately,

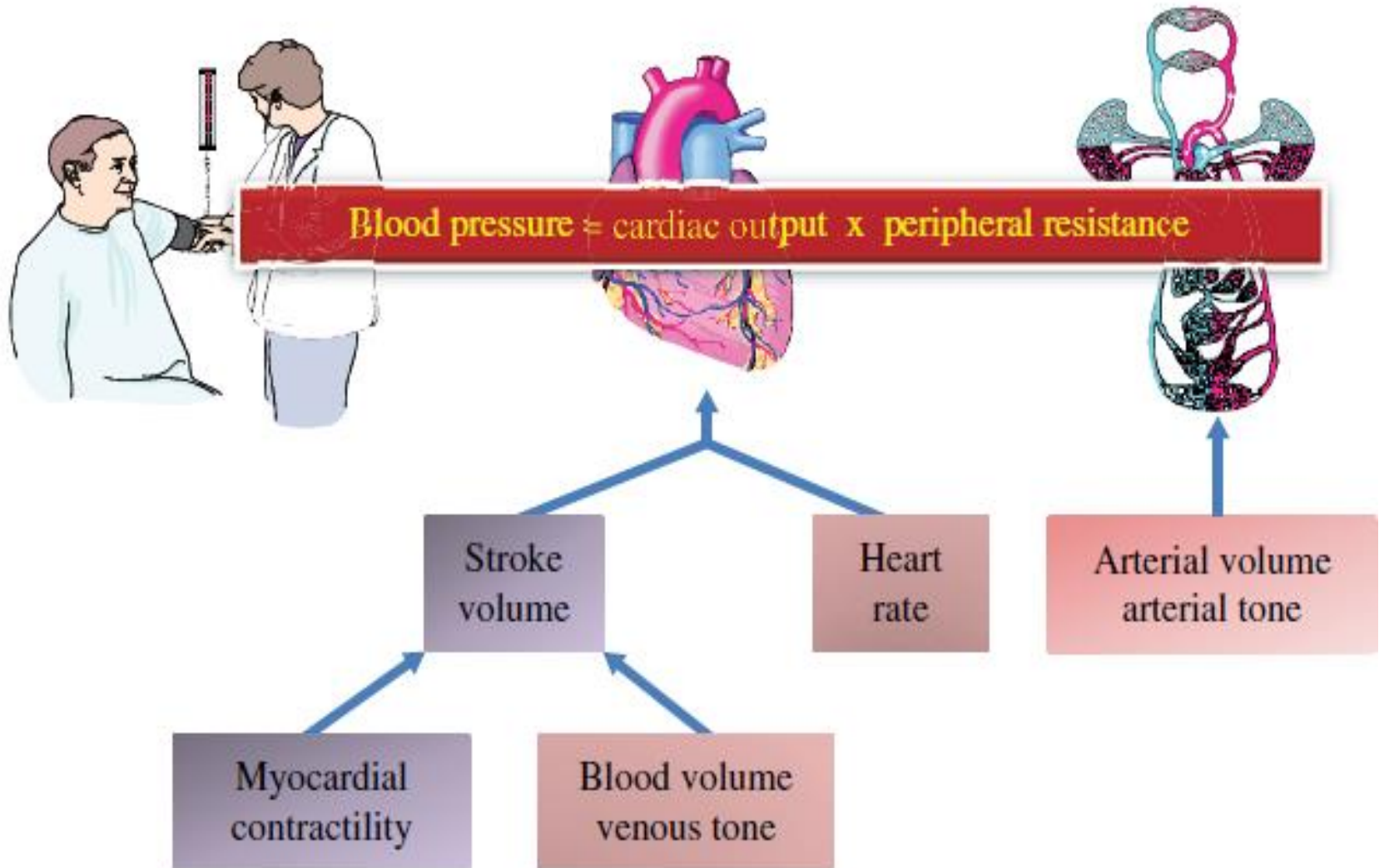
Hypertension can lead to:

- ✓ Myocardial infarction
- ✓ Stroke
- ✓ Heart failure
- ✓ Renal failure, and death

Categories of BP in Adults

Classification	SBP (mm Hg)	Vs	DBP(mm Hg)
Normal	<120	and	<80
Elevated	120-129	or	<80
Hypertension			
Stage 1	130-139	or	80-89
Stage 2	≥140	or	≥90

BP regulation



BP regulation...

1) Baroreflexes adjust moment-to-moment blood pressure

- ✓ Baroreceptors in the aortic arch and carotid sinuses
 - A decrease in BP activates the baroreceptor reflex → activates sympathetic nervous system.

2) The RAAS provides tonic, longer term regulation of blood pressure.

- ✓ Decreased renal perfusion pressure stimulates **renin release** and activation of RAAS (constriction & aldosterone release).

Pathophysiology of hypertension development

- Major pathophysiological mechanisms of hypertension include:
 - ✓ Activation of the sympathetic nervous system
 - ✓ Activation of the RAAS
 - ✓ Endothelial dysfunction ($\downarrow$ NO & $\uparrow$ endothelin)
 - ✓ Oxidative stress ($\uparrow$ ROS)

Non-pharmacologic interventions to reduce BP include:

- ✓ weight loss
- ✓ sodium restriction
- ✓ ↑ dietary potassium intake
- ✓ increase physical activity
- ✓ decrease alcohol intake
- ✓ decrease saturated fat and total fat

Antihypertensive Drugs

- Antihypertensive agents are drugs which reduce blood pressure via
 - either decreasing the CO or peripheral resistance.
- **Classification of antihypertensive drugs:**
 1. Sympatholytic drugs
 - a. Centrally acting Sympatholytics
 - a. Central α_2 agonists: clonidine, α -methyldopa
 - b. Autonomic ganglia blocking drugs
 - Hexamethonium
 - c. Adrenergic neuron-blocking agents:
 - Guanethidine, Guanadrel, Reserpine

Antihypertensive agents.....

d. adrenergic receptor blockers

- a. Alpha-blockers: Prazosin, terazosin, doxazosin, phentolamine
- b. Beta-blocker: propranolol, atenolol, Metoprolol
- c. Nonselective alpha & beta blockers: Carvidolol, labetalol

2. Drugs acting on vascular smooth muscle

a. Arteriolar vasodilators

- i. Calcium channel blockers
- ii. ii. Hydralazine
- iii. Minoxidil

b. Arteriolar & venular vasodilation

E.g. Sodium nitroprusside

Antihypertensive agents.....

3. Drug acting on Renin-angiotensin-aldosterone system

- Drugs which block renin release; E.g. Beta-blockers
- Direct renin inhibitors: Aliskerin
- Angiotensin converting enzyme inhibitors; E.g. Captopril, enalapril
- Angiotensin II receptor antagonists; E.g. losartan
- Aldosterone antagonist; E.g. spironolactone

4. Diuretics

➤ E.g. Thiazides

Clonidine

- **MOA:** stimulate α_2 -adrenoceptors in the brain



Activate an inhibitory neuron



Reduce sympathetic out flow from CNS



↓ Peripheral Resistance, ↓CO



↓ BP

Clonidine.....

Therapeutic use:

1. Management of mild to moderate hypertension (alone or in combination)
2. Opiates, alcohol, nicotine withdrawal symptom
3. Diagnostic agent for a patient suffered from pheochromocytoma
 - Clonidine ↓NE release only in normal individual
- Oral dose should be discontinued gradually over 2-4 days
 - To avoid rebound hypertension (receptor up-regulation).

Clonidine.....

Adverse effect

- Drowsiness
- Dizziness
- dryness of mouth
- GI disturbance
- sexual dysfunction (impotence)

α - Methyldopa

Mechanism of action

- α - methyldopa converted to α -methyl noradrenaline.
- decreases sympathetic outflow by acting on α_2 -receptor

Pharmacological Action

- Hypotension - $\downarrow$ CO, $\downarrow$ PR
- Hypotensive effect is potentiated when it is combined with thiazide diuretics.
- **Methyldopa is preferably used during pregnancy.**

α -Methyldopa.....

Therapeutic use

- moderate to severe hypertension
 - 250mg-1000mg po bid, increase every 2 days as needed (max dose - 3gm/day)

Adverse effect

- Headache
- Fatigue
- forgetfulness
- Mental depression
- sleep disturbance

Hydralazine

Mechanism of action

- act on arteriolar smooth muscle & prevent the vasoconstrictor effect of NA.

Therapeutic use

- moderate to severe hypertension
- Congestive heart failure

Adverse effect

- **Tachycardia**
- sweating
- nasal congestion
- flushing
- lupus erytheromatus-like syndrome (with high doses)

Minoxidil

Mechanism of action

- K^+ channel activator, results in arteriolar relaxation
- ↑ed cutaneous blood flow results in stimulation of resting hair follicle

Therapeutic uses

- Moderate to severe hypertension
- For male pattern baldness (used topically).
 - the drug doesn't cause formation of hair follicle.
 - It only cause elongation of hair follicle.

Minoxidil.....

Adverse effects:

- reflex tachycardia
- Hypertrichosis (↑ hair growth)
- Headache
- sweating
- H₂O/Na⁺ retention & edema
 - must be used in combination with a β blocker and a loop diuretic

CALCIUM CHANNEL BLOCKERS

- Verapamil, diltiazem
- Dihydropyridine family (amlodipine, felodipine, isradipine, nicardipine, nifedipine, and nisoldipine)
 - *Verapamil* is the least selective of any CCB and has significant effect on CC of both
 - ✓ cardiac and vascular smooth muscle cells.
 - Like *verapamil*, *diltiazem* affects both cardiac and vascular smooth muscle cells, but it has a less pronounced negative inotropic effect on the heart compared to that of *verapamil*
 - All dihydropyridines have a much greater affinity for vascular calcium channels than for calcium channels in the heart.
 - ✓ particularly beneficial in treating hypertension.

Calcium Channel blockers...

- **Pharmacological actions**

1. Negative inotropic effect

- Decrease myocardial contractility
- Decrease workload and oxygen consumption
 - **Hence used for treatment of angina**

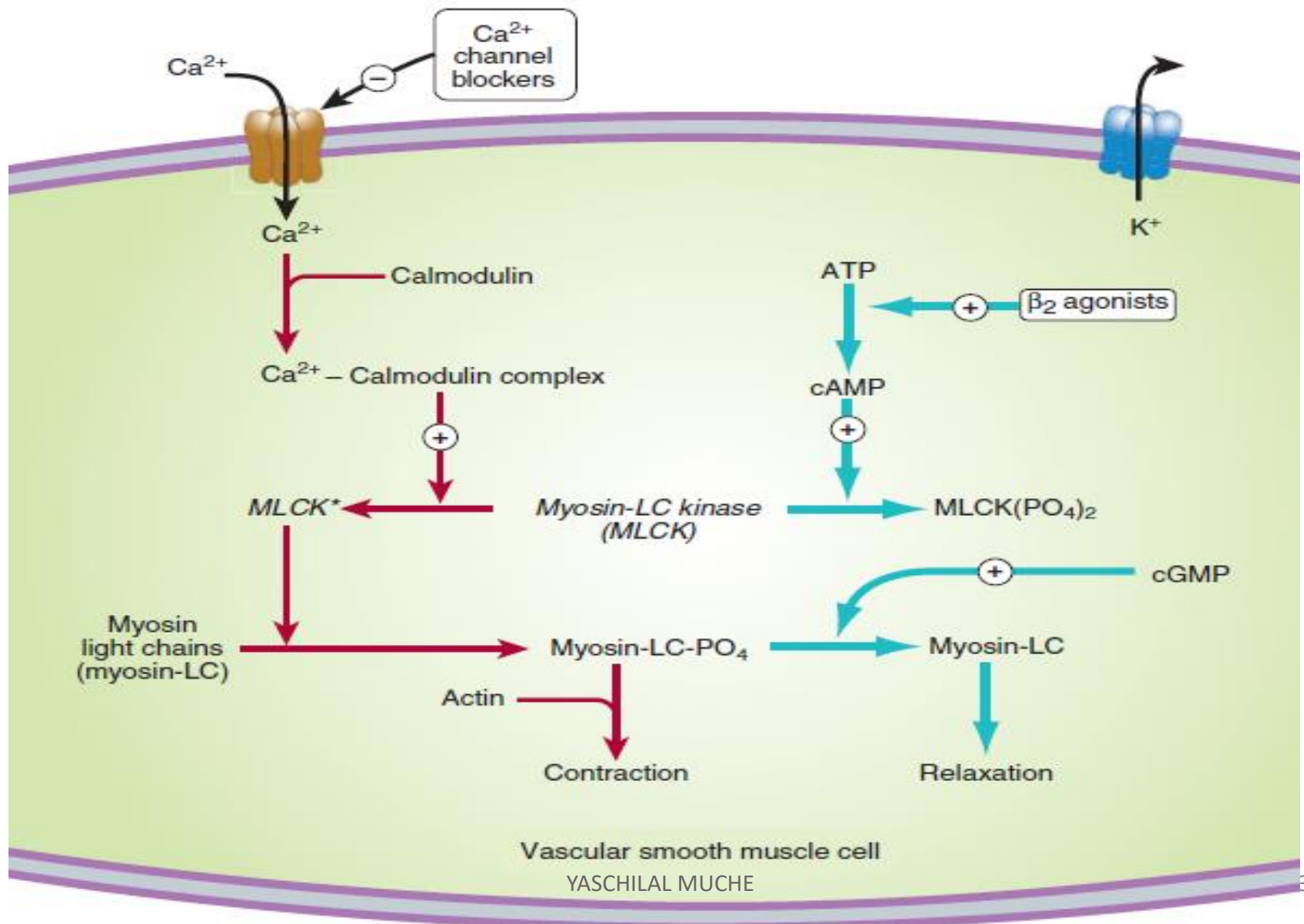
2. Anti-arrhythmic effect

- Suppress ectopic pacemaker activity
 - Ectopic pacemaker are those which are pace maker other than the original (SA node) like bundles.

3. Decrease Blood pressure

- Relaxation of vascular smooth muscle
 - Among 1st line drugs in mild-moderate hypertension
 - can be used in hypertensive emergency

Control of smooth muscle contraction and site of action of calcium channel-blocking drugs;



Therapeutic uses of CCBs

- HTN
 - CCBs may be used as an initial therapy or as add-on therapy.
- All CCBs are useful in the treatment of angina.
- In addition, *diltiazem and verapamil* are used
 - in the treatment of atrial fibrillation.

Sodium nitroprusside

- Arteriolar and venular vasodilator
- Disadvantage:
 - short duration of action hence should be given IV by continuous infusion.
- Light sensitive:
 - it is converted to cyanide when exposed to light, hence bottle should be wrapped by brown or black paper.

Adverse effect:

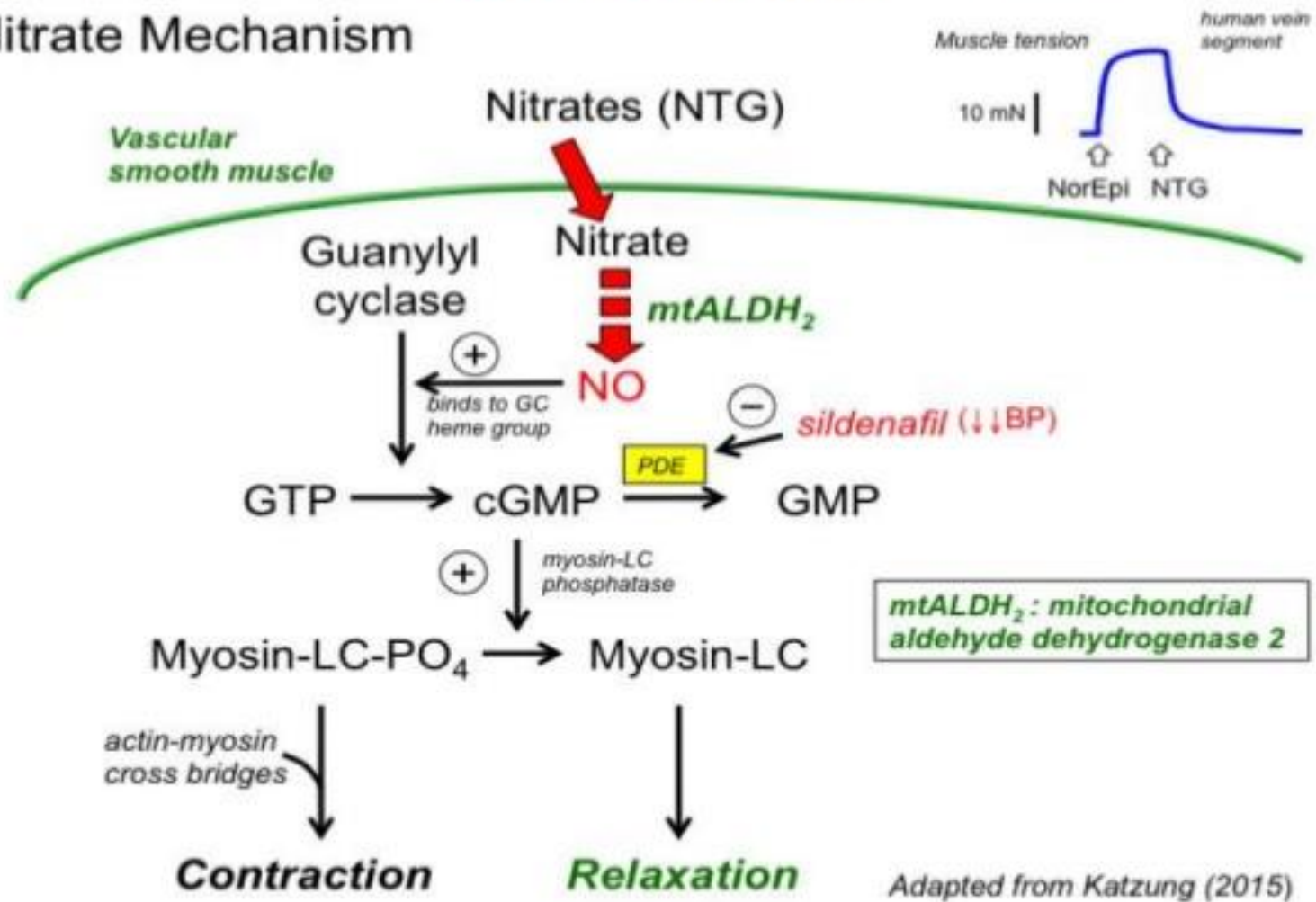
- Liver dysfunction

TABLE 11–3 Mechanisms of action of vasodilators.

Mechanism	Examples
Release of nitric oxide from drug or endothelium	Nitroprusside, hydralazine, nitrates, ¹ histamine, acetylcholine
Reduction of calcium influx	Verapamil, diltiazem, nifedipine
Hyperpolarization of smooth muscle membrane through opening of potassium channels	Minoxidil, diazoxide
Activation of dopamine receptors	Fenoldopam

Mechanism of action

Nitrate Mechanism



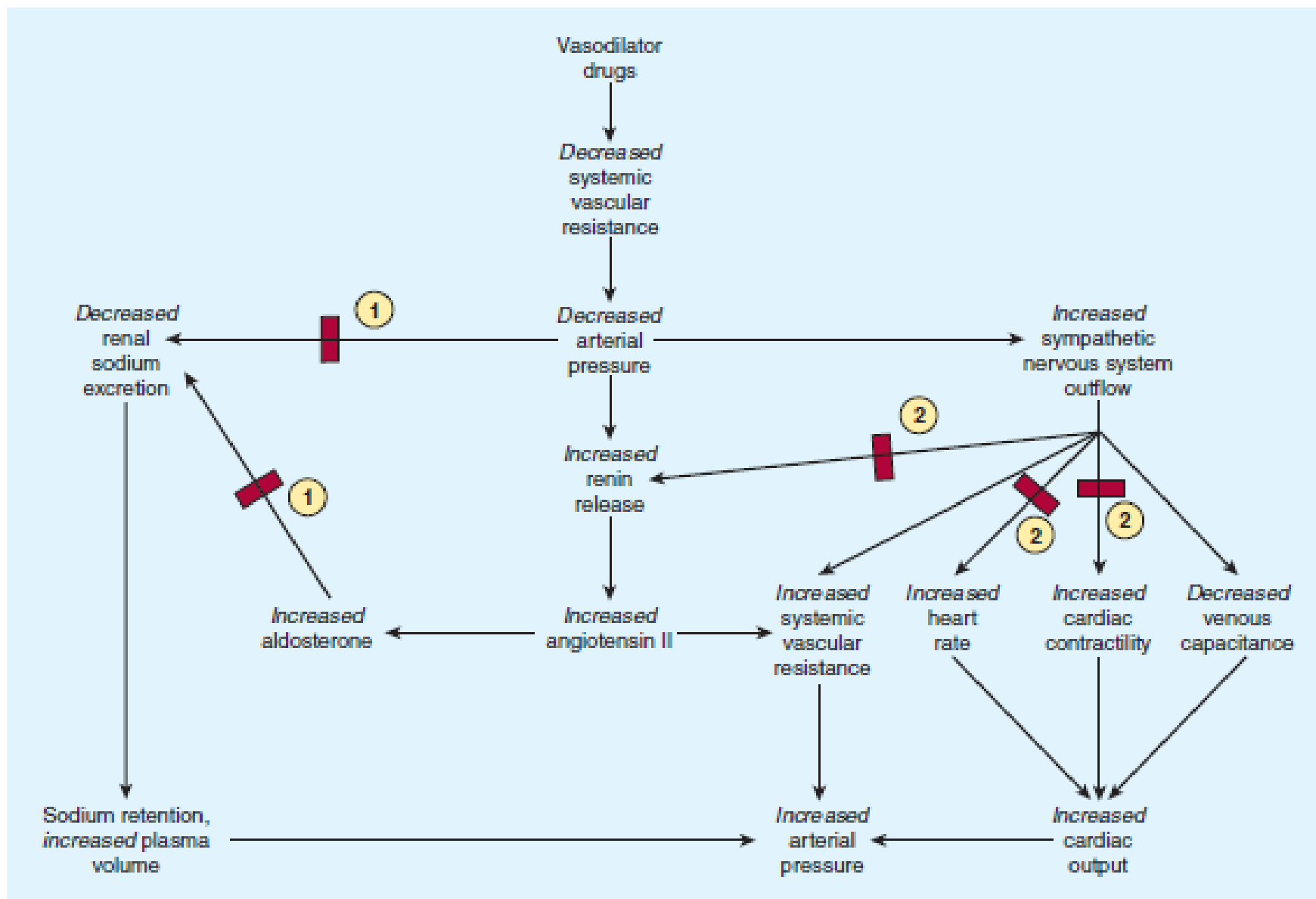


FIGURE 11-4 Compensatory responses to vasodilators; basis for combination therapy with β blockers and diuretics. ① Effect blocked by diuretics. ② Effect blocked by β blockers.

Drugs acting on RAAS

Angiotensin II result in:

- Direct Vasoconstriction → ↑ Peripheral resistance
- Activation of SNS → ↑ CO & Peripheral resistance
- Secretion of aldosterone ---> retention of sodium & water ---> ↑ blood volume - ↑ CO

Antihypertensive drugs acting on RAAS:

- B-blockers inhibit release of renin
- renin inhibitor
- Angiotensin converting enzyme inhibitors (ACEIs)
- Angiotensin II receptor antagonist

Renin inhibitors

- A selective renin inhibitor, **aliskiren**,
 - is available for the treatment of hypertension.
- It acts earlier in the RAAS than ACEIs or ARBs
- As with ACEIs and ARBs,
 - aliskiren is contraindicated during pregnancy

ACEIs

- Among 1st line antihypertensive drugs
- hypertensive emergencies (enalaprilat)

Contraindications of ACEIs:

- PREGNANCY (Risk category D)
 - Fetotoxic if taken at any time during pregnancy
- Bilateral renal artery stenosis
- severe aortic stenosis
- renal insufficiency
- hyperkalemic states
- Past history of angioedema

ARBs

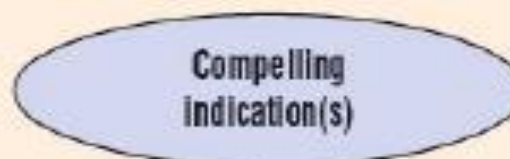
- Are among first line antihypertensive drugs

Contraindications of ARBs:

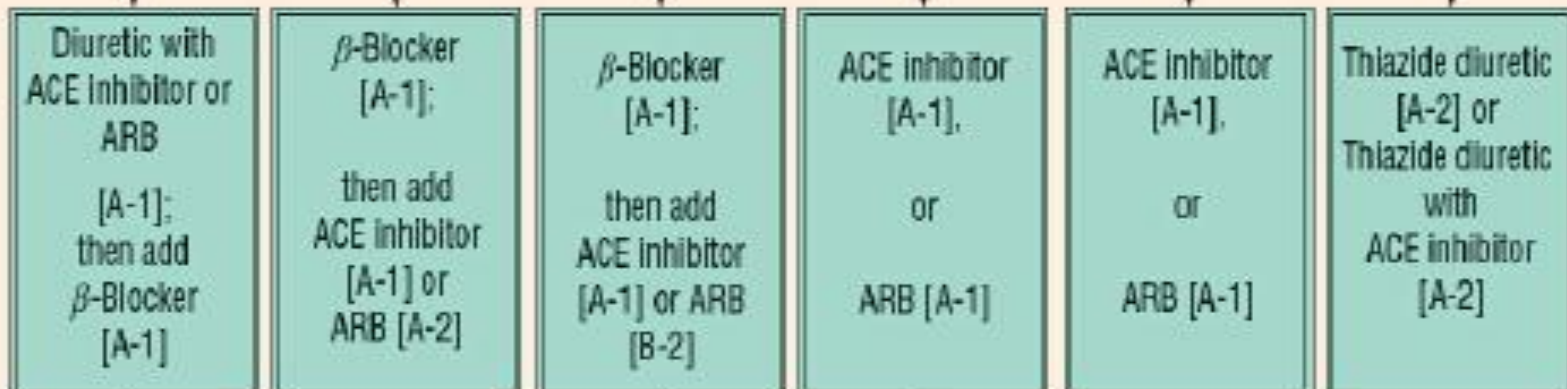
1. PREGNANCY (Risk category D)
2. Bilateral renal artery stenosis
3. severe aortic stenosis
4. renal insufficiency

1st line drugs for hypertension with no comorbidity

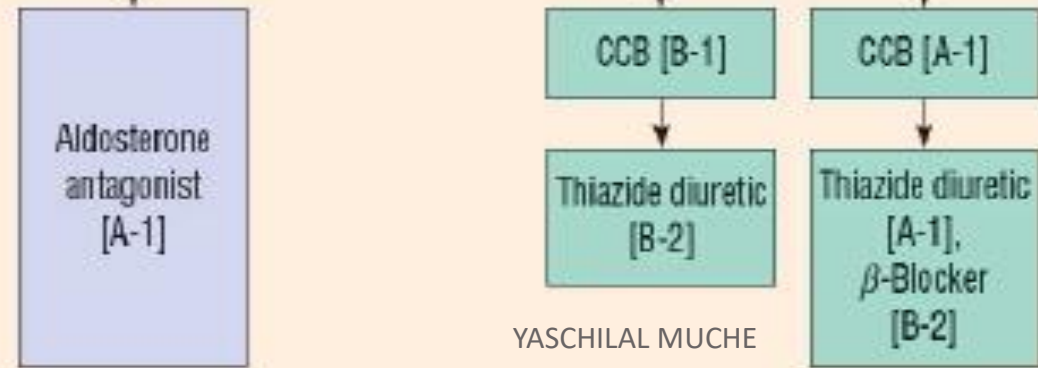
- ACEIs
- ARBs
- CCBs
- Thiazide diuretics



Standard Pharmacotherapy



Add-on Pharmacotherapy



Drug therapy of angina pectoris

Drug therapy of angina pectoris (Ischemic heart disease)

- Angina pectoris is not a disease rather it is a symptom
- severe chest pain occurs due to
 - imbalance between oxygen supply and oxygen demand of myocardium (ischemic heart disease)
- Etiology – coronary artery spasm, atherosclerosis or blood clott.
- Anti-anginal drugs may relieve attacks of acute myocardial ischemia;
 - By increasing myocardial oxygen supply or
 - by decreasing myocardial oxygen demand or
 - both

Types of angina pectoris

A. Stable angina, effort-induced angina, classic or typical angina

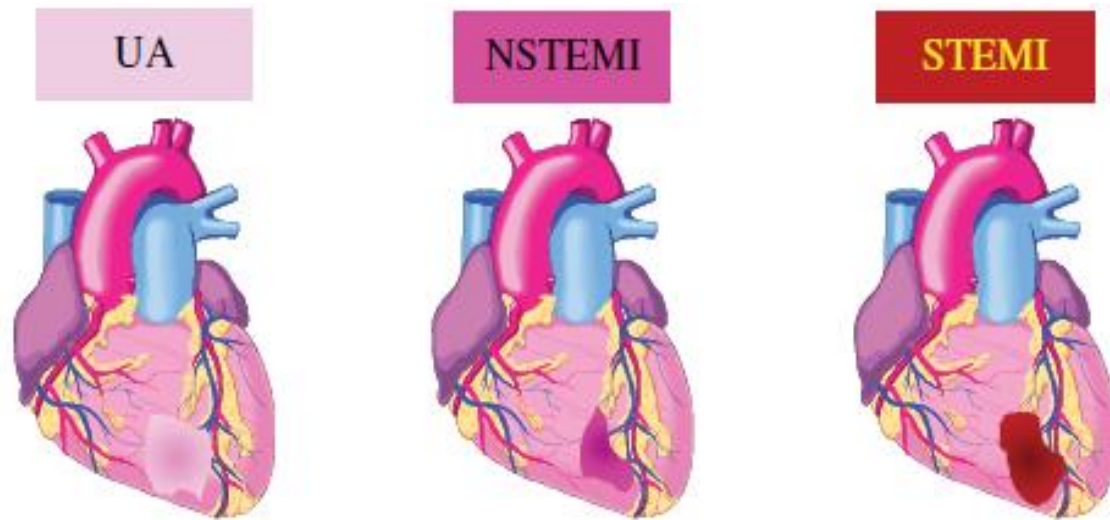
- Stable angina is characterized by predictable pain on exertion.
 - Cause is atherosclerosis
 - Occur when there is high oxygen demand
 - Exercise, eating, coitus
 - Subside when oxygen demand is withdrawn

Types of angina.....

B. Unstable angina

- Unstable angina is classified between stable angina and MI.
- episodes of angina occur at rest and when
 - there is an increase in the severity, frequency, and duration of chest pain in patients with previous stable angina.
- Unstable angina is a form of **acute coronary syndrome**
- requires hospital admission and more aggressive therapy to prevent progression to MI and death.
 - sublingual **NTG, ASA, and an anticoagulant** (UFH, enoxaparin, fondaparinux, or bivalirudin).
 - intranasal oxygen (if oxygen saturation is low)

Characteristics of acute coronary syndromes:



Ischemia	++	+++	++++
Cell death	–	+	++
Occlusion	Incomplete	Incomplete	Complete
ST elevation	No	No	Yes

Types of angina.....

C. Prinzmetal, variant, vasospastic, or rest angina

- occur as a result of ***vasospasm*** of coronary vessels.
- Although individuals with this form of angina may have significant coronary atherosclerosis,
 - the angina attacks are unrelated to physical activity, heart rate, or blood pressure.
- Prinzmetal angina generally responds promptly to coronary vasodilators, such as
 - *nitroglycerin and calcium channel blockers.*

Antianginal drugs

- *decrease myocardial oxygen requirement* by decreasing the determinants of oxygen demand
 - organic nitrates
 - calcium channel blockers
 - β blockers
- increase myocardial oxygen delivery by reversing coronary arterial spasm.
 - nitrates
 - calcium channel blockers

Organic Nitrates

- **Nitroglycerin**
- **Isosorbide mononitrate**
- **Isosorbide dinitrate**
- With the exception of nitroglycerin, which is a liquid having a high vapor pressure,
 - these compounds are solid at room temperature.
- All organic nitrates are highly lipid soluble.

Mechanism of Action

- lead to the formation of the reactive free radical NO.

Pharmacological action

- Platelet- Inhibit platelet aggregation
- Smooth muscle (bronchi, gastrointestinal, uterine, urethral smooth muscle) – relaxation
- Blood vessels
 - Dilation of veins; Preload reduction
 - Dilation of arteries; After load reduction
 - Dilation of Coronary Artery; increased O₂ supply to the heart

Nitroglycerine

- Use:
 - antianginal agent
 - Pulmonary hypertension
- Dose: Sublingual - 1mg TID
 - Transdermal patch: 0.2 - 0.4mg/hour

Adverse effect

- Vasodilation (Flushing, Headache, Fainting and Sweating)
- Tolerance - Develop with time for preload, afterload and coronary vasodilation
 - Develop when nitrates are continuously administered

Calcium Channel-Blocking Drugs

- block voltage-dependent calcium channels in vascular smooth muscle and the heart
 - Decrease the afterload by decreasing vascular resistance
 - Increase O₂ supply to the myocardium due to relaxation of the coronary arteries

Use

- chronic treatment of stable angina
- prophylaxis of coronary vasospasm or variant angina

B-blockers

- Although they are not vasodilators, β -blocking drugs are extremely useful in the management of
 - Stable angina (angina associated with effort).

Effects;

- *decreased heart rate & contractility* → decrease myocardial oxygen requirements at rest and during exercise
- **Use- stable angina**

Drug therapy of chronic heart failure

Introduction

- Heart failure (HF) is a complex, progressive disorder in which the heart is unable to pump sufficient blood to meet the needs of the body.
- caused by;
 - systolic dysfunction (**SHF or HFrEF**),
 - diastolic dysfunction (**DHF or HFpEF**), or both.
- Causes of Systolic dysfunction;
 - Reduction in muscle mass (e.g., myocardial infarction)
 - Dilated cardiomyopathies
 - Ventricular hypertrophy
- Causes of Diastolic dysfunction;
 - Increased ventricular stiffness
 - Mitral or tricuspid valve stenosis and
 - Pericardial disease

PATHOPHYSIOLOGY

- Impaired ability of the heart to contract and/or relax results in a decrease in CO.
- The decrease in CO results in activation of compensatory responses to maintain the circulation.
 - Tachycardia and increased contractility (**activation of SNS**)
 - Na⁺ and water retention (**activation of RAAS**)
 - vasoconstriction (**NE, angiotensin II, ET-1, neuropeptide Y, urotensin II, and AVP**)
 - Ventricular hypertrophy and remodeling (**due to chronic activation of the first three compensatory MZMs**)→**decreased CO**

Stage A

Patients at high risk for developing HF



Development of structural heart disease

Stage B

Patients with structural heart disease but no HF signs or symptoms



HF symptoms develop

Stage C

Patients with structural heart disease and current or previous symptoms



Rx-resistant symptoms

Stage D

Refractory HF requiring specialized interventions

Common Examples

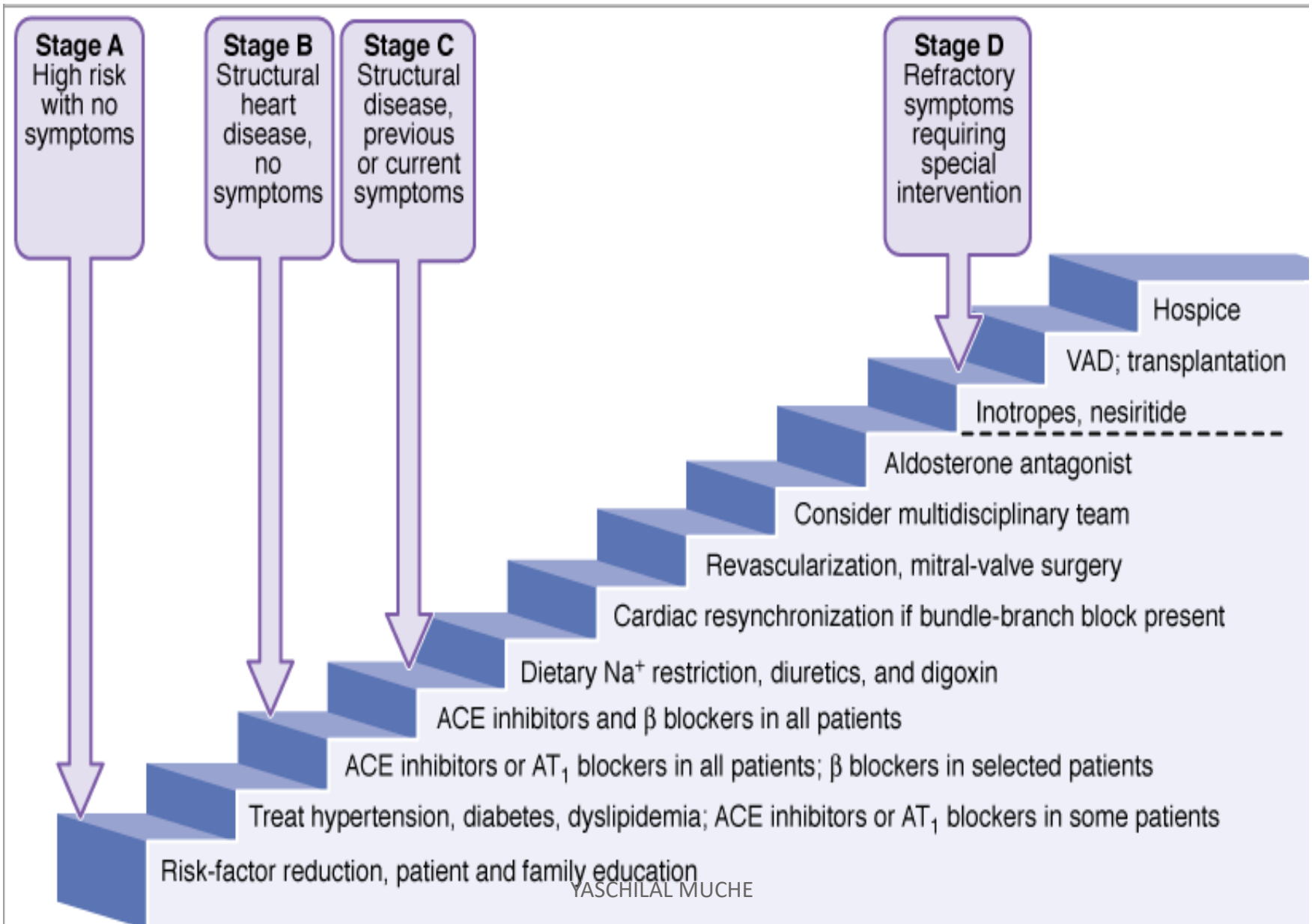
HTN, CAD or other atherosclerotic vascular disease, DM, obesity, metabolic syndrome.

Previous MI, left ventricular hypertrophy, left ventricular systolic dysfunction

Left ventricular systolic dysfunction and symptoms such as dyspnea, fatigue, and reduced exercise tolerance.

Patients requiring recurrent hospitalization or who cannot be discharged without mechanical assist devices or inotropic therapy

Pharmacotherapy of heart failure



ACEIs and ARBs

- reduce peripheral resistance and thereby reduce afterload.
- reduce salt and water retention (by reducing aldosterone secretion) ---> reduce preload.
- ACE inhibitors improve
 - clinical signs and symptoms of HF and
 - have been shown to significantly **improve patient survival in HF**
- Angiotensin receptor blockers should be considered in patients intolerant to ACE inhibitors.

β-blockers

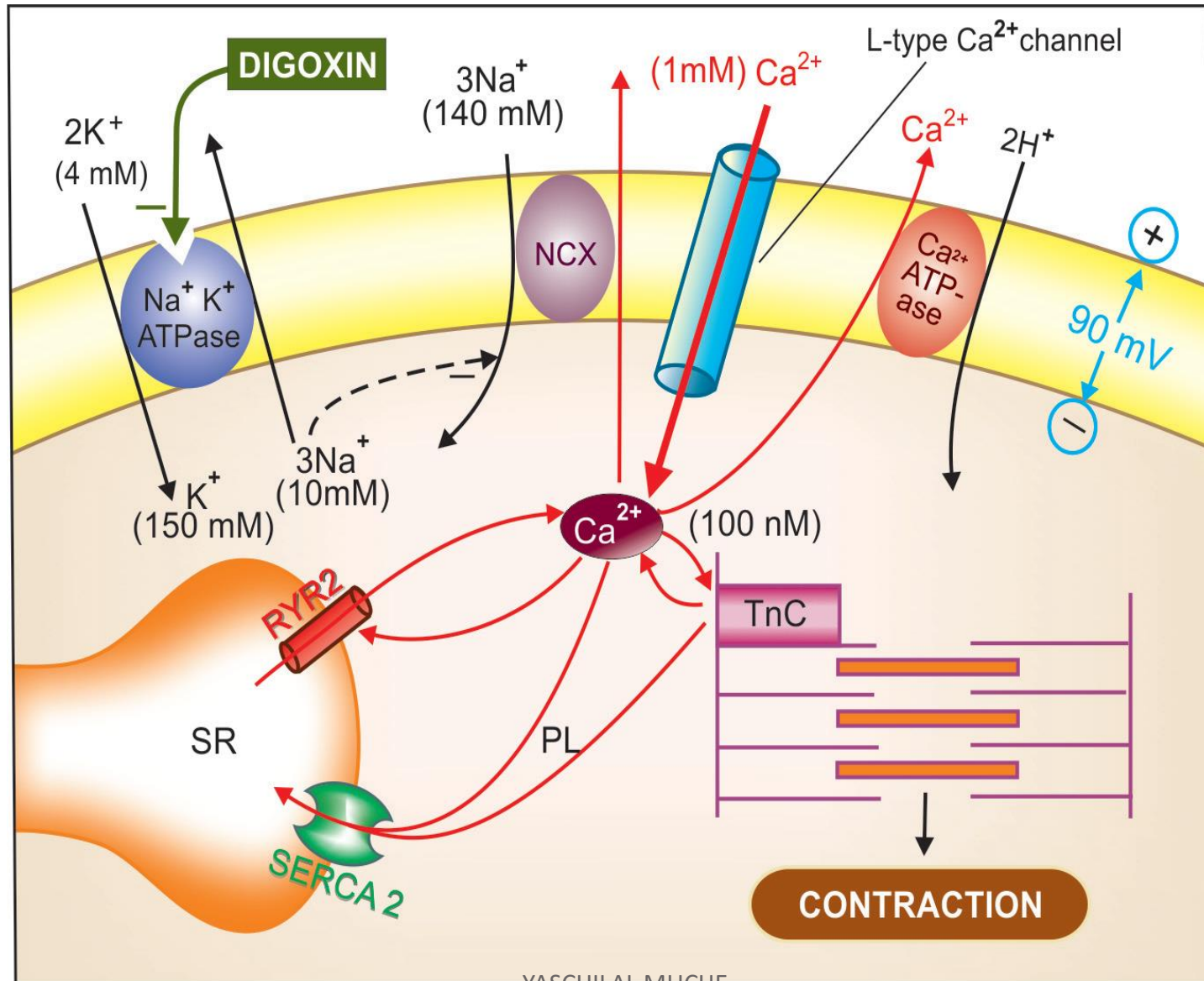
- **Carvedilol, Metoprolol succinate (CR/XL), and Bisoprolol**
 - are the BBs shown to have a benefit in the pharmacotherapy of CHF
- In patients with SHF, BBs
 - decrease hospitalizations
 - **slow disease progression and improve survival**
 - improve quality of life in many patients with HF, although this is not a universal finding

Inotropic drugs

A. Cardiac Glycosides

- positive inotropic effect on failing myocardium.
 - Cardiac glycosides increase contraction of the cardiac sarcomere by increasing the free calcium concentration.
 - MOA; inhibit the $\text{Na}^+\text{-K}^+$ -ATPase pump, which causes an increase in intracellular Na^+ , slowing the rate of the $\text{Na}^+/\text{Ca}^{2+}$ -exchanger, and thereby causing an increase in intracellular Ca^{2+}
 - bind to K^+ binding site at the pump, and inhibit the exchange

Mechanism of positive inotropic action of cardiac glycosides



Pharmacological action;

1. Cardiac action

A. Contractility

- $\uparrow \text{FC} \longrightarrow \uparrow \text{blood supply to the heart} \longrightarrow \downarrow \text{diastolic size of the heart} \longrightarrow \downarrow \text{energy consumption}$
 - A digitalized heart can do more work with the same energy or same work with less energy.

B. Heart rate

- Digitalis decrease HR - due to its vagal action

C. Automaticity

- Ability of cardiac tissue to generate impulse by themselves is increased

D. Coronary circulation

- $\uparrow$ coronary blood flow due to $\uparrow \text{CO}$

Pharmacological action

E. Digitalis and K^+

- Hypokalemia increase the action of digitalis.
 - normal dose in hypokalemic patient could induce toxicity.

F. Digitalis and Ca^{++}

- Ca^{++} increases force of contraction of the heart like digitalis.
- Hence Excess of Ca^{++} with digitalis may produce synergistic effect.
 - This will again produce digitalis toxicity.

2. Extra cardiac action

- a. Diuretic effect - By increasing renal blood flow
- b. GIT- Nausea, Vomiting

Dose: 0.125 - 0.5mg po per day

0.1 - 0.4mg IV or Im

Digitalis toxicity

- Digitalis have a narrow therapeutic index

A. cardiac arrhythmia

- Disturbed impulse formation due to ectopic pace maker activity
 - ventricular arrhythmia which may lead to death.

B. GIT - NV

C. Skin rashes, gynecomastia, neuralgic pain in face and extremities, vertigo.

Treatment of digitalis toxicity:

1. Stop digitalis therapy
 2. Stop diuretic therapy- Most diuretic produce hypokalemia
 3. Treat bradycardia with atropine
 4. Mild toxicity can be treated with K^+
- Avoid Quinidine and verapamil because they $\uparrow$ digitalis level
 - An antidote, digoxin immune Fab (DIGIBIND, DIGIFAB), is available to treat toxicity

B. β -Adrenergic agonists

- **Dobutamine and dopamine**, improve cardiac performance by causing positive inotropic effects and vasodilation.
- β -Adrenergic agonists lead to an increase in intracellular cAMP, which results in the activation of protein kinase A.
 - Protein kinase A then phosphorylates slow calcium channels, thereby **increasing entry of calcium ions** into the myocardial cells and enhancing contraction
- *Dobutamine is the* most commonly used inotropic agent other than *digoxin*.
- Both drugs must be given by intravenous infusion and are primarily ***used in the short-term treatment of acute HF in the hospital setting.***

Vasodilators

- If the patient is intolerant of ACE inhibitors or β -blockers, or if additional vasodilator response is required,
 - a combination of *hydralazine* and *isosorbide dinitrate* may be used.
- A fixed-dose combination of these agents has been shown to
 - improve symptoms and **survival in black patients** with SHF (HFrEF) on standard HF treatment (β -blocker plus ACE inhibitor or ARB).

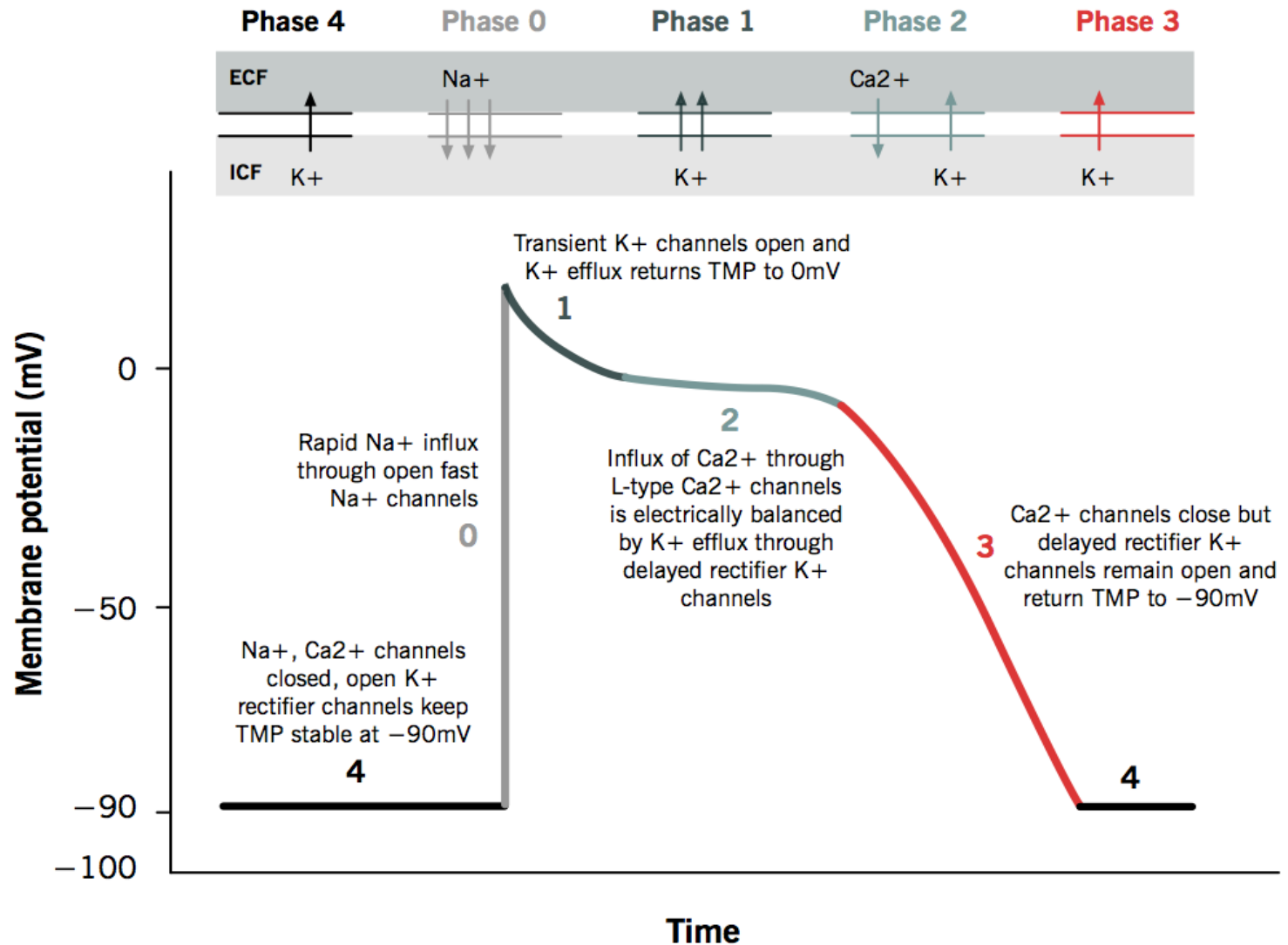
Antiarrhythmic Drugs

Introduction

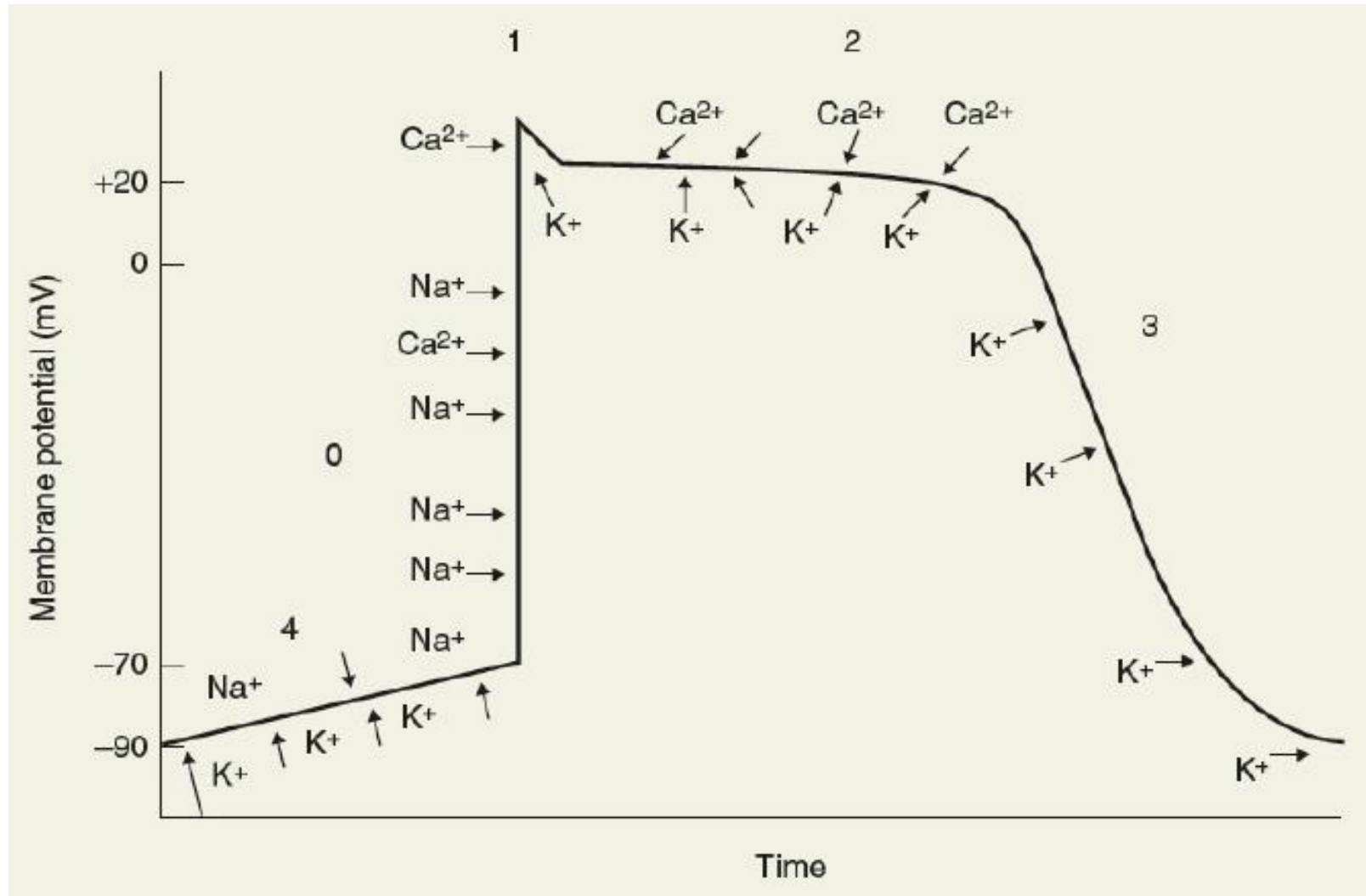
- ✓ Arrhythmia is abnormal rhythm of the heart.
- ✓ The normal heart rhythm includes:
 - i. HR: (60-100 beat/min)
 - ii. Origin of electrical activity: SA node (sinus rhythm)
 - iii. Normal conduction path way $\Rightarrow$ SA $\rightarrow$ Atrium $\rightarrow$ AV $\rightarrow$ Bundle of his & Purkinje $\rightarrow$ Ventricles
 - iv. Normal velocity
 - Atria: modest, 1 m/s
 - AV: very slow, 0.05 m/s
 - Bundle of his & purkinje: fast, 4-5 m/s
- Hence, any deviation from the normal rhythm can cause arrhythmia

Action potential of cardiac muscles

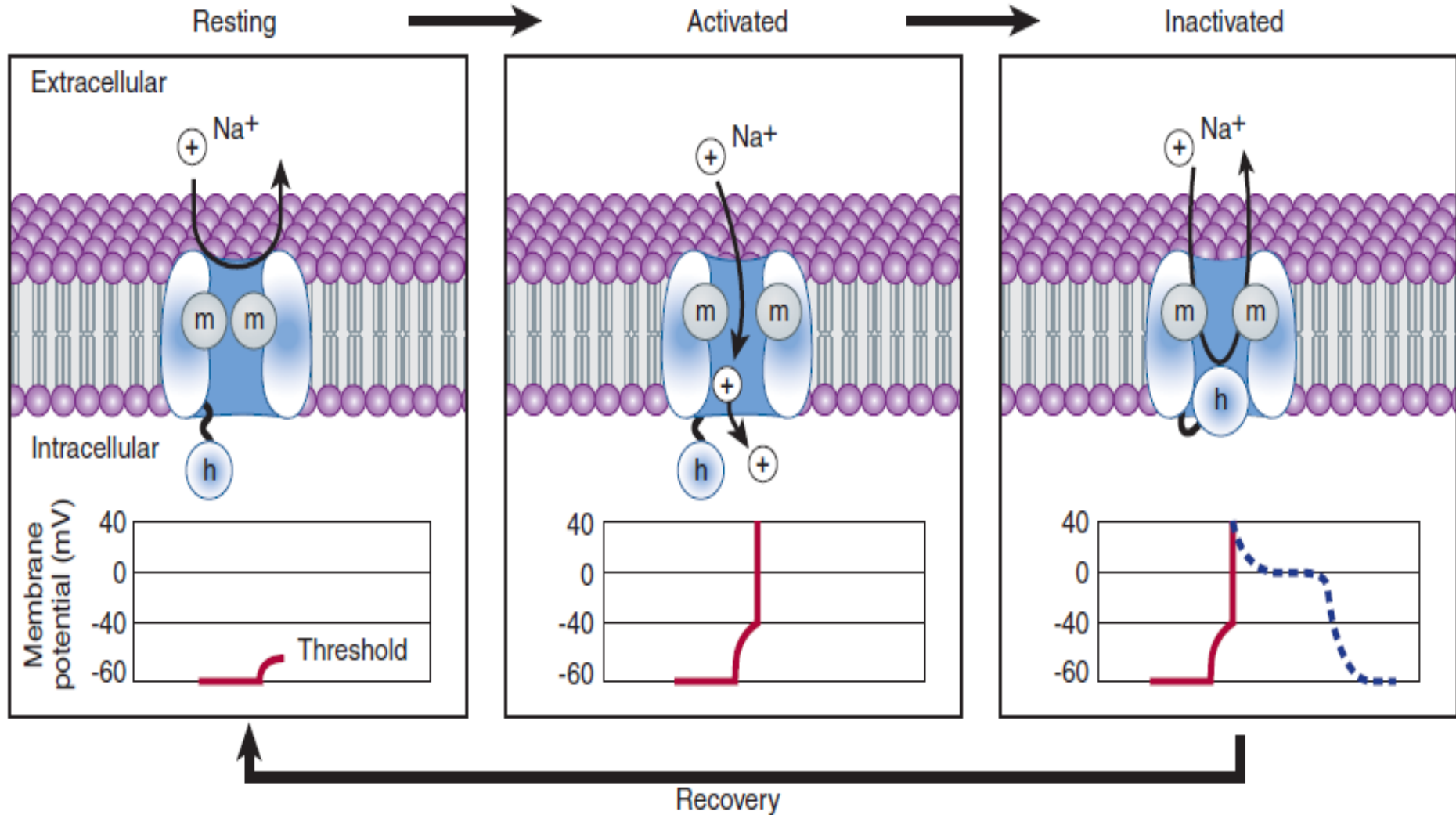
Grigoriy Ikonnikov and Eric Wong



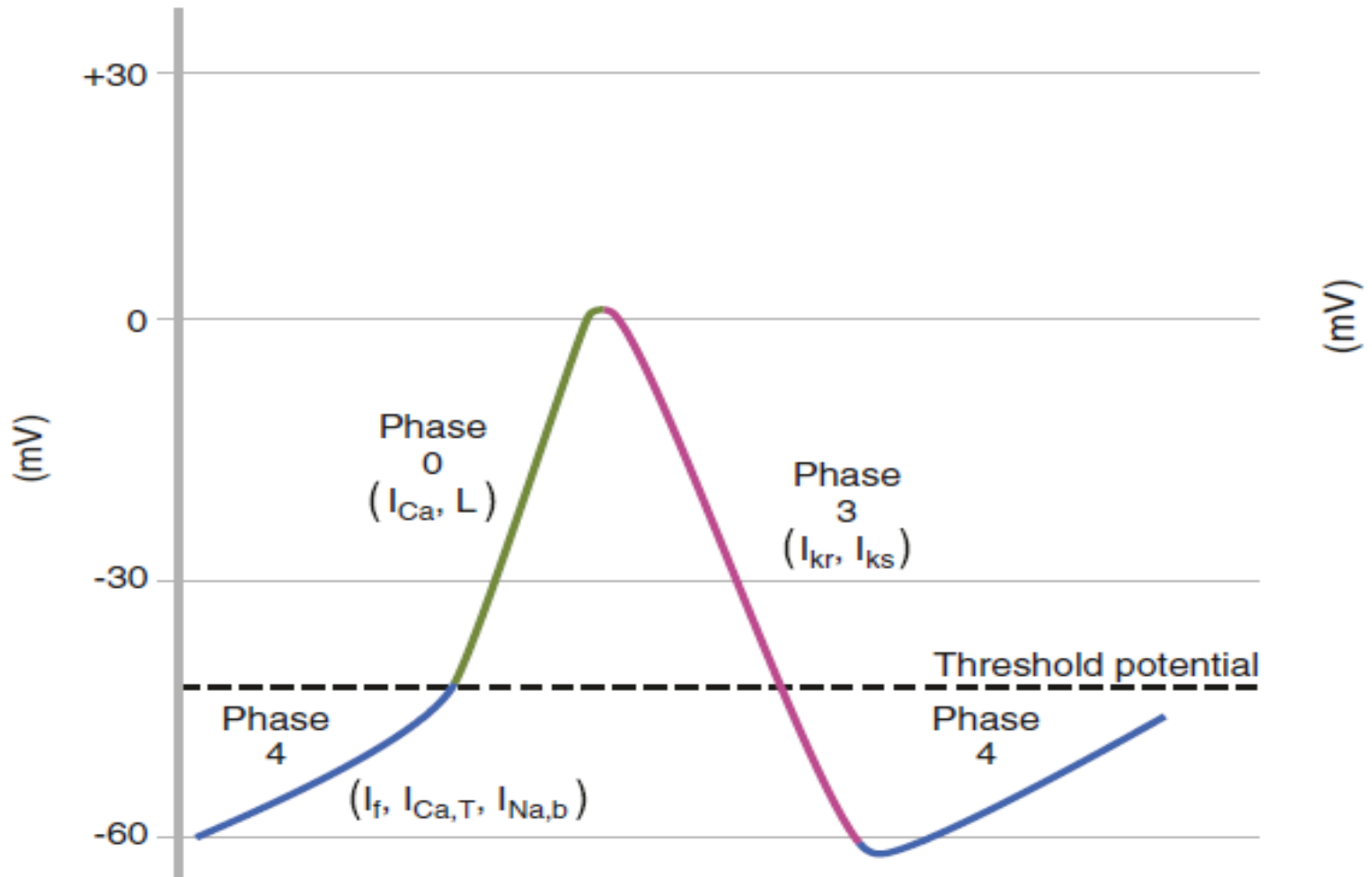
Purkinje fiber action potential showing specific ion flux responsible for the change in membrane potential;

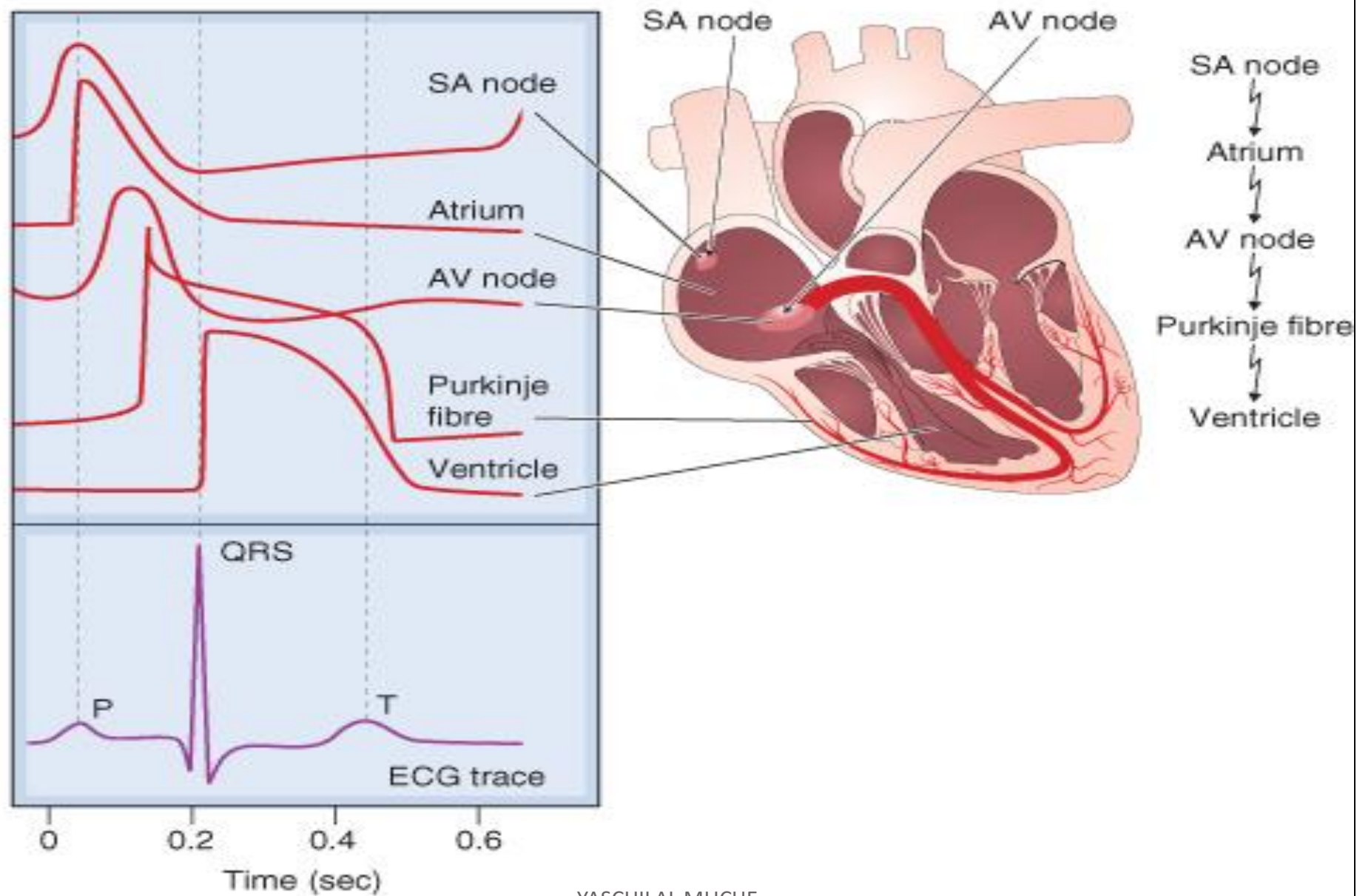


Channel states of Na⁺ current



Slow-Response Fibers (SA and AV Nodes)



B

Cardiac arrhythmias

Types of tachy arrhythmias

- Simple tachycardia: 100-150 beat/min
- Paroxysmal tachycardia: 150-250 beat/min
- Flutter: 250-350 beats/min
- Fibrillation: >350 beats/min

Brady arrhythmias

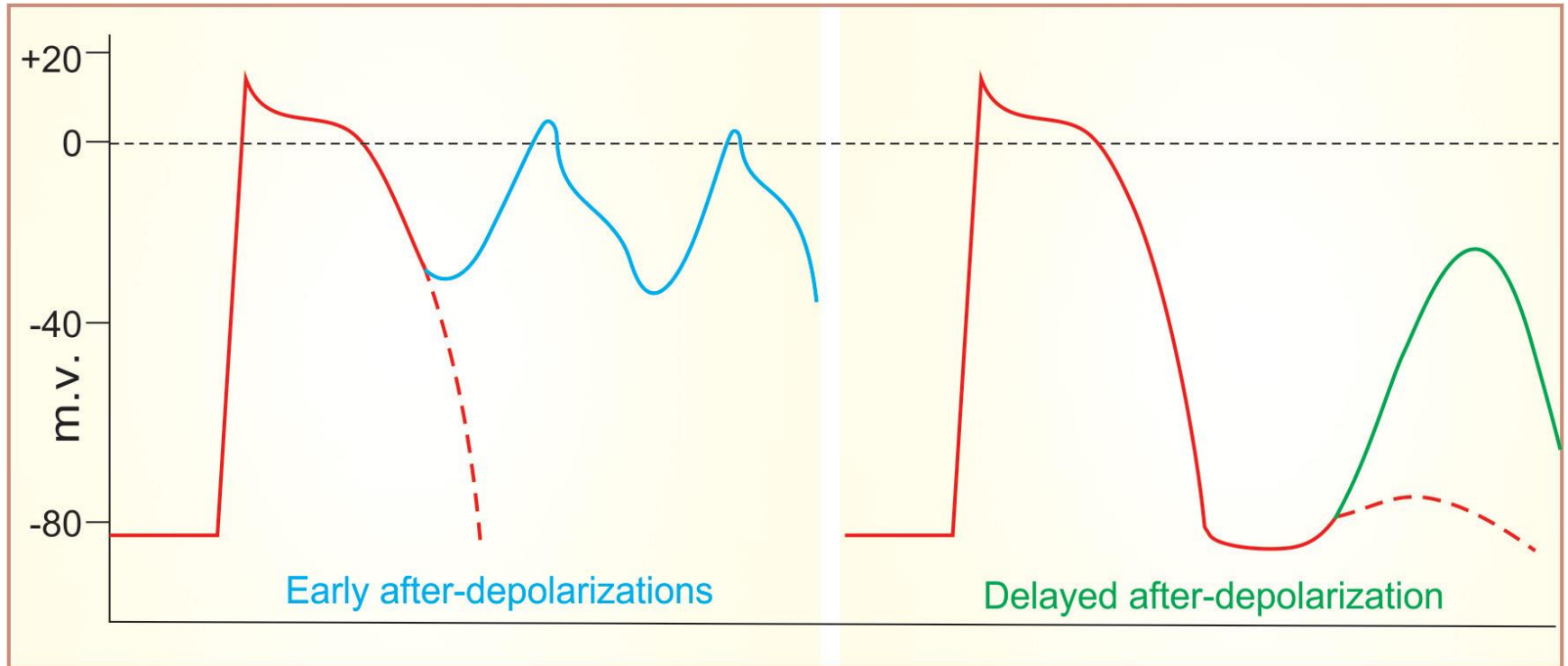
- Sinus bradycardia
- Nodal bradycardia/heart block

Mechanisms of Arrhythmias

1) Abnormality in impulse formation (automaticity)

- ❖ Slow normal pacemaker rate
- ❖ Acceleration of normal pacemaker rate
- ❖ Triggered automaticity
 - Afterdepolarizations
 - ✓ transient depolarizations that interrupt phase 3 (**early afterdepolarizations, EADs**) or phase 4 (**delayed afterdepolarizations, DADs**).

EADs & DADs:



2) Abnormality in impulse conduction

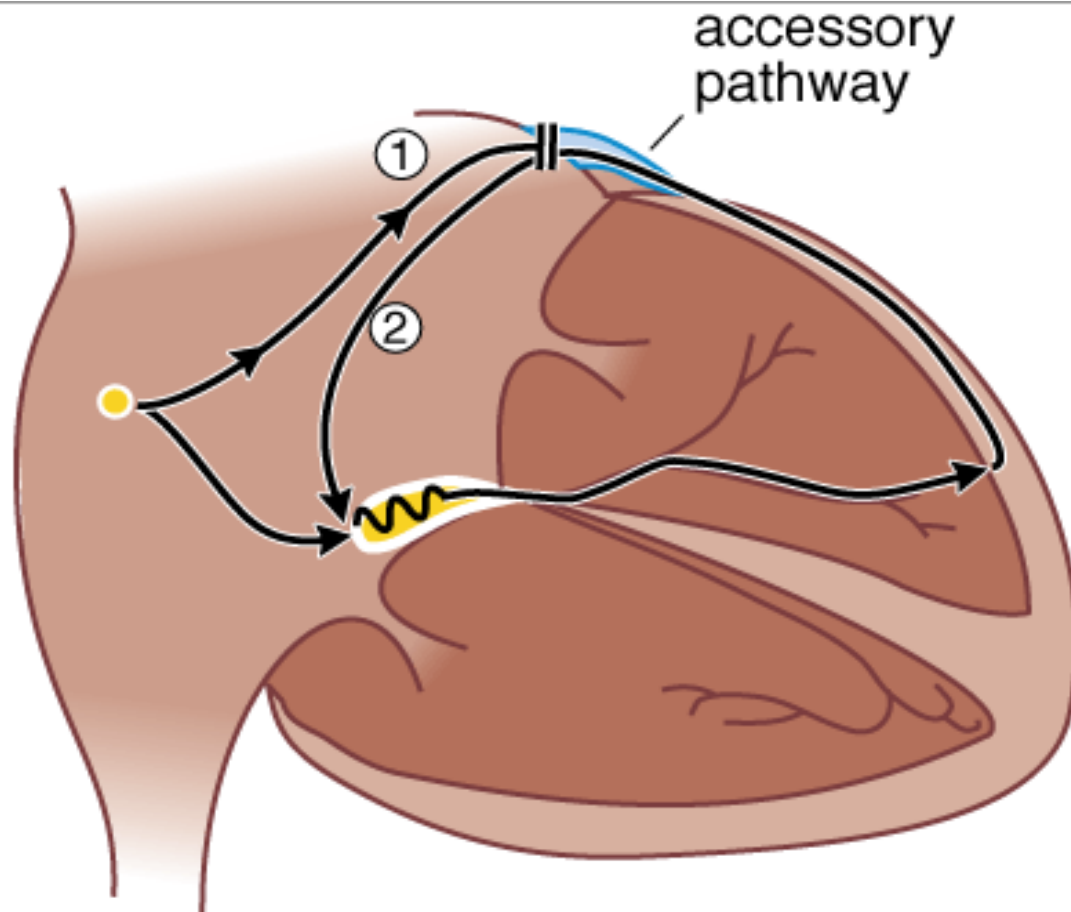
- ❖ Impulse block: eg, AV block

- ❖ Reentry (Circus movement)

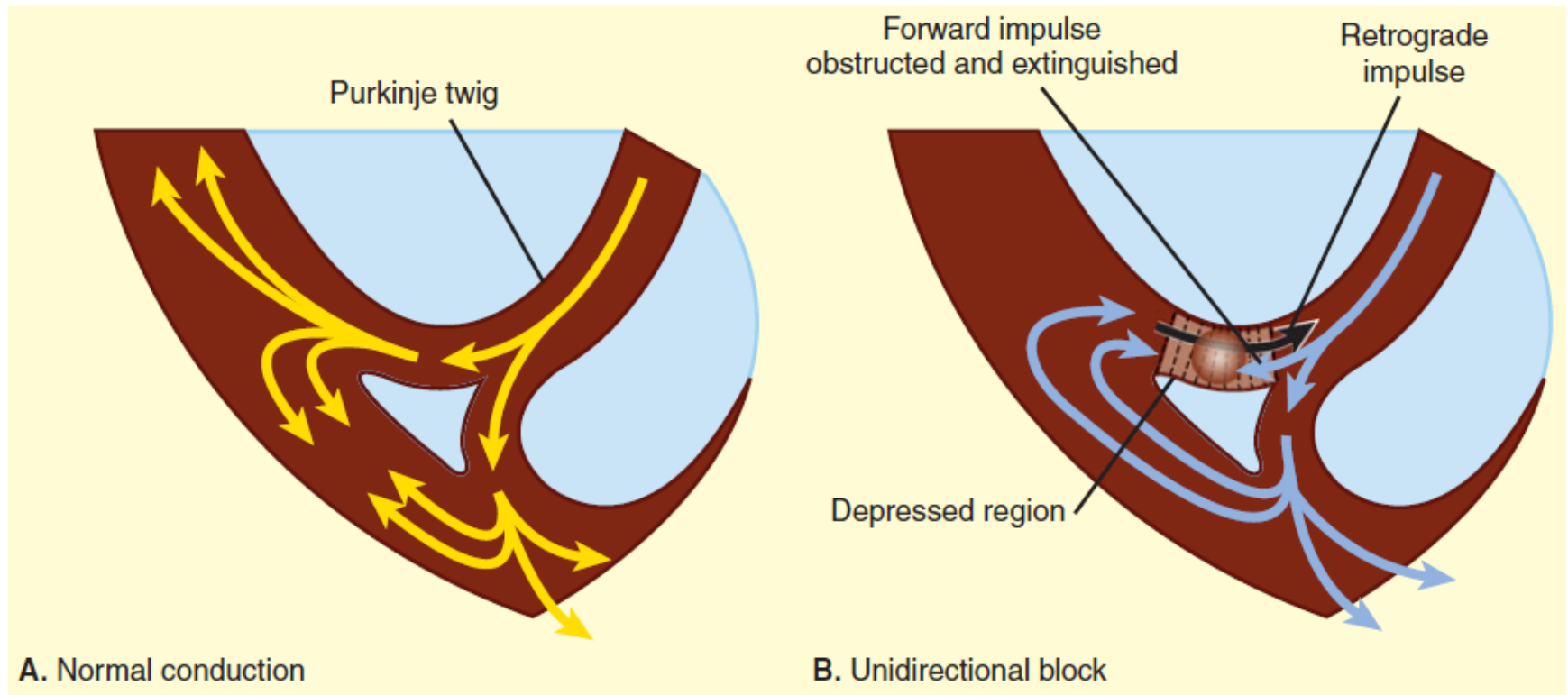
- **anatomically defined**; for example, in Wolff-Parkinson-White syndrome, the reentry circuit consists of atrial tissue, the AV node, ventricular tissue, and an accessory AV connection (bundle of Kent, a bypass tract).

- **Functionally Defined**; For example, a premature beat (from an EAD or DAD) from within the ventricular wall may experience refractory tissue in only one direction, allowing for conduction throughout the rest of the wall until the originally refractory area recovers, re-excites, and then propagates back through the original location of the premature beat.

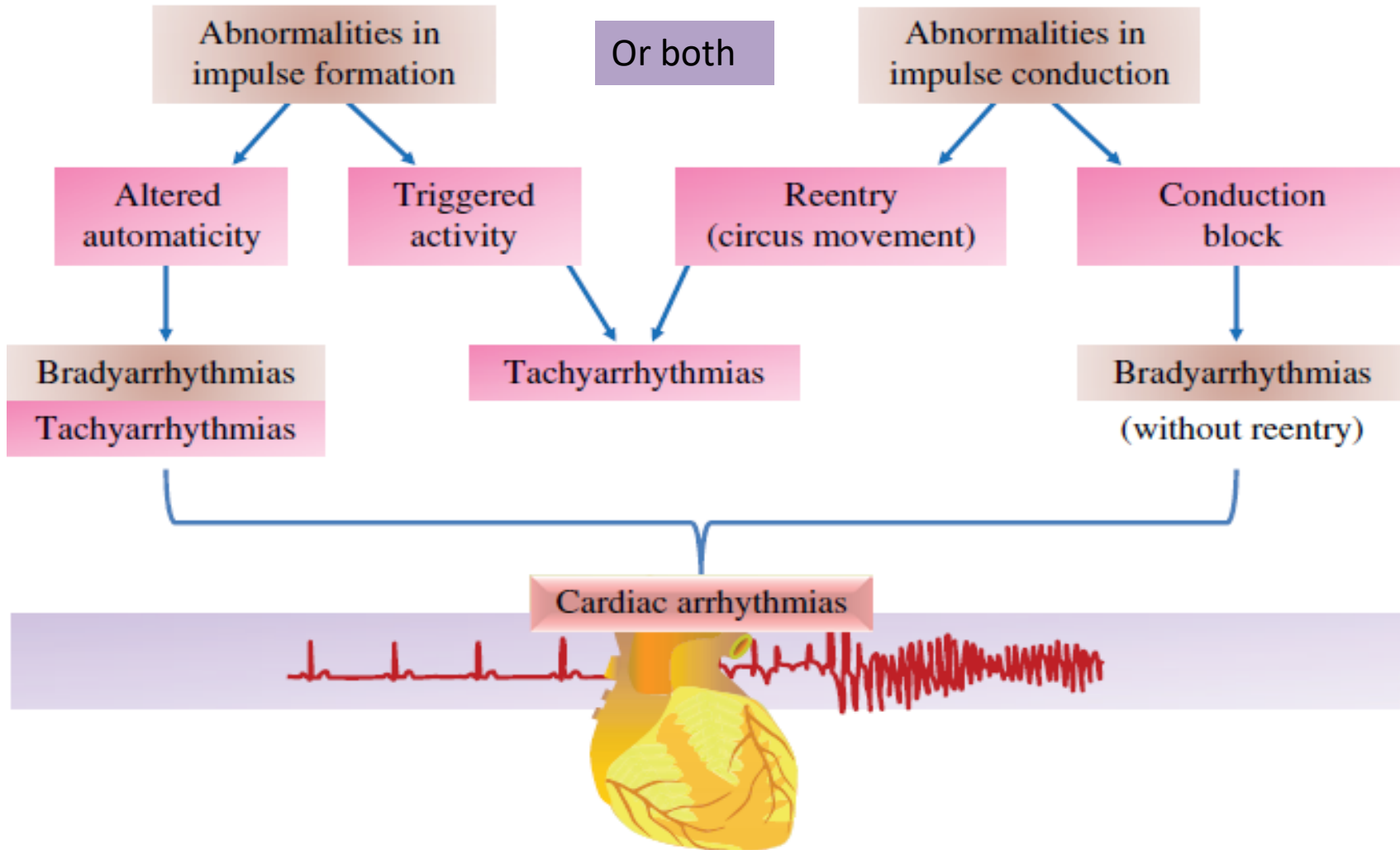
Atrioventricular re-entrant tachycardia in the Wolff-Parkinson-White syndrome.



Schematic diagram of a reentry circuit that might occur in small bifurcating branches of the Purkinje system where they enter the ventricular wall.



General mechanisms of cardiac arrhythmias



Cause for arrhythmia

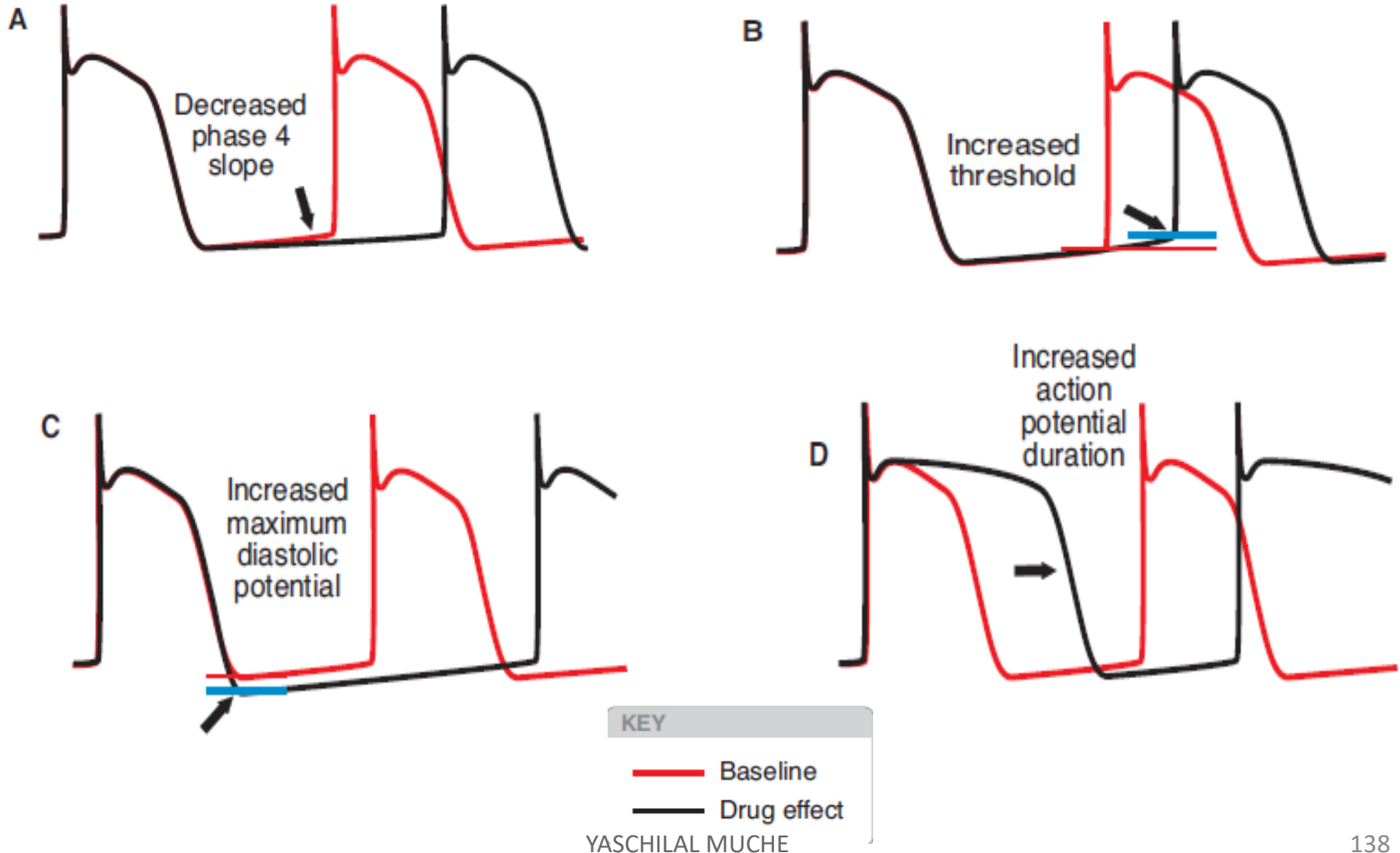
- Electrolyte and PH imbalance
 - hypokalemia
- Mechanical injury
- neurogenic
- drug
 - digoxin
- Ischemia

Treatment strategies

✓ Drugs for altered/enhanced spontaneous automaticity

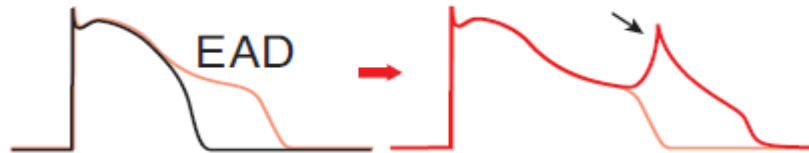
- a) Decrease phase 4 slope (NaCB, CCB, BB)
- b) Increase threshold potential (NaCB or CCB)
- c) ↓RMP & decrease slope (vagus stimulation & adenosine)
- d) Increase action potential duration (KCB): prolonged repolarization & total number of AP decreased.

Treatment strategies...



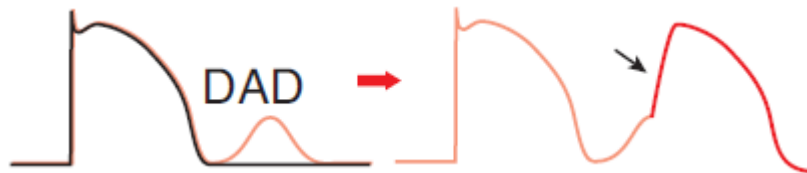
Treatment strategies ...

✓ Drugs for triggered activity



- Reduce duration of action potential by $\uparrow$ HR (hence, repolarization is completed before Na^+ channels are activated)

☞ Isoproterenol, Mg^{++} & pacing of heart



- CCBs ($\downarrow \text{Ca}^{++}$ over load) and NaCB ($\uparrow \text{TP}$)

Treatment strategies ...

✓ Drugs for abnormal conduction

- For functionally defined tachy arrhythmia
 - ✓ Increase effective refractory period by NaCB or KCB (fast response tissue) & slow response tissue (CCB)
- For WPWS re entry
 - ✓ Inhibit the accessory pathway (NaCBs) or
 - ✓ Inhibit AV node conduction & ↑refractoriness
 - CCB, BBs, adenosine, digoxin

Antiarrhythmic Drugs

- There are different classes of antiarrhythmic drugs
 - Class I: Na⁺ channel blocker
 - Quinidine, Procainamide, Lidocaine, Mexiletine, Phenytoin, Flecainide
 - Class II: Beta-adrenoceptor antagonist
 - Metoprolol, Propranolol, Esmolol
 - Class III: K⁺ channel blocker
 - Amiodarone, Ibutilide, Sotalol
 - Class IV: Ca⁺ channel blocker
 - Verapamil, Diltiazem
 - Others- Adenosine, Magnesium sulfate, Digoxin, potassium.

Class IA

- Block fast Na⁺ channels
 - Preferentially in the open or activated state
 - state dependent blockage
- Increase APD and ERP
- Also blocks K⁺ channel (prolongs repolarization)

Quinidine

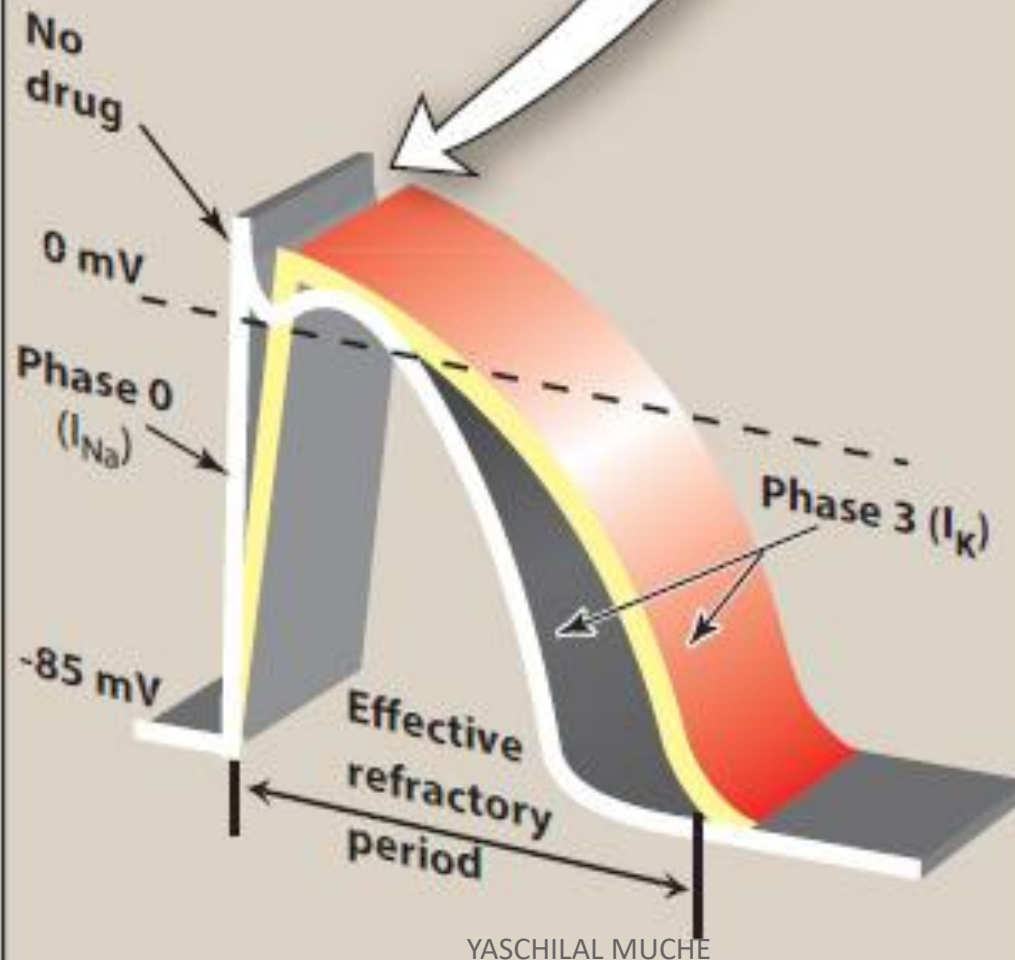
- In addition to the above, causes muscarinic receptor blockage, which can **increase HR and AV conduction**
- May also cause **vasodilation** via alpha blockage with possible reflex tachycardia
- Orally effective, wide clinical use in many arrhythmias;
 - in atrial fibrillation, need **initial digitalization** to slow AV conduction

- AE: cinchonism (GI disturbance, tinnitus, ocular dysfunction, CNS excitation), hypotension, prolongation of QRS and increased QT interval associated with syncope (**torsades**).
- DI: hyperkalemia enhances effects and vice versa; displaces digoxin from tissue binding sites, enhancing toxicity.

Procainamide

- Less muscarinic receptor blockage
- Metabolized via N-acetyltransferase to N-acetyl procainamide (NAPA), an active metabolite
- AE: systemic lupus erythematosus (SLE)-like syndrome more likely with slow acetylators; hematotoxicity (thrombocytopenia, agranulocytosis); CV effect (torsades)

Class IA drugs slow Phase 0 depolarization. In addition, because of their Class III activity, these drugs prolong the action potential.



Class IB

- Block fast Na⁺ channels
- Block inactivated channels – preference for tissues partly depolarized (slow conduction in hypoxic and ischemic tissues)
 - increased threshold for excitation and less excitability of hypoxic muscle
- **Decrease APD** – due to blockage of the slow Na⁺ “window” currents,
- **Decrease ERP**

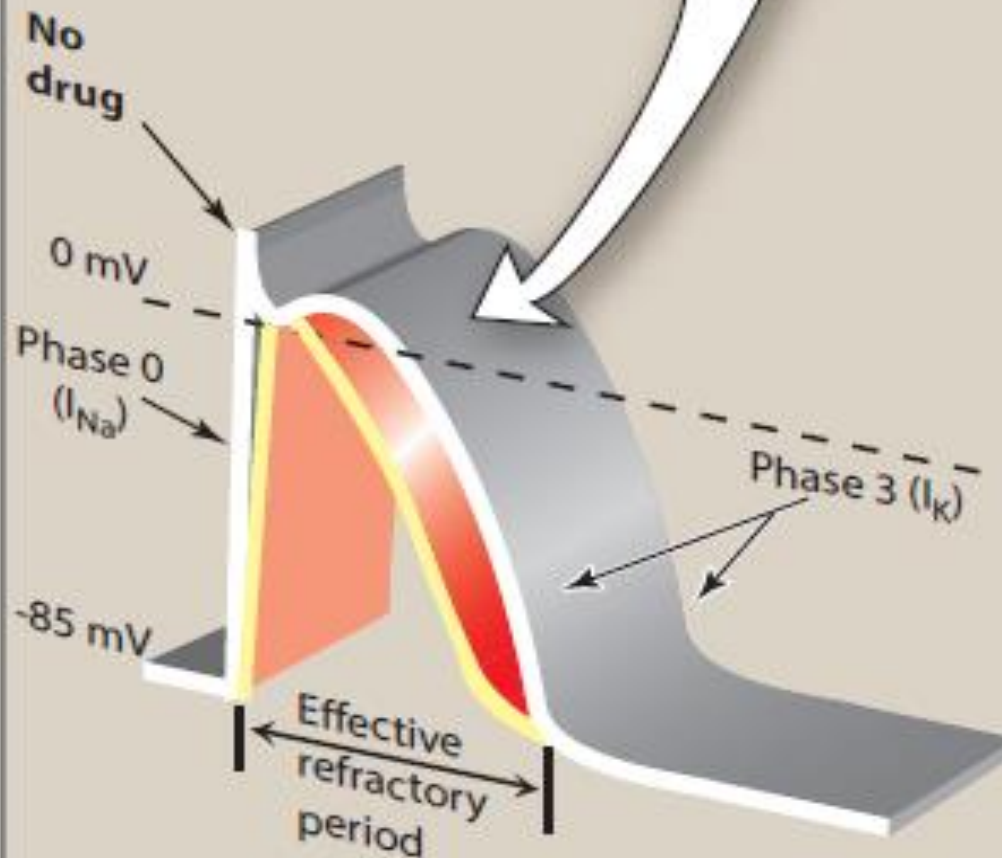
Lidocaine

- Uses: post-MI, open-heart surgery, digoxin toxicity
- SE: CNS toxicity (seizures), least cardiotoxic of conventional antiarrhythmics
- IV use because of first-pass metabolism

Mexiletine and tocainide

- Same use as lidocaine
- Oral formulations

**Class IB drugs shorten
Phase 3 repolarization
and decrease the duration
of the action potential.**

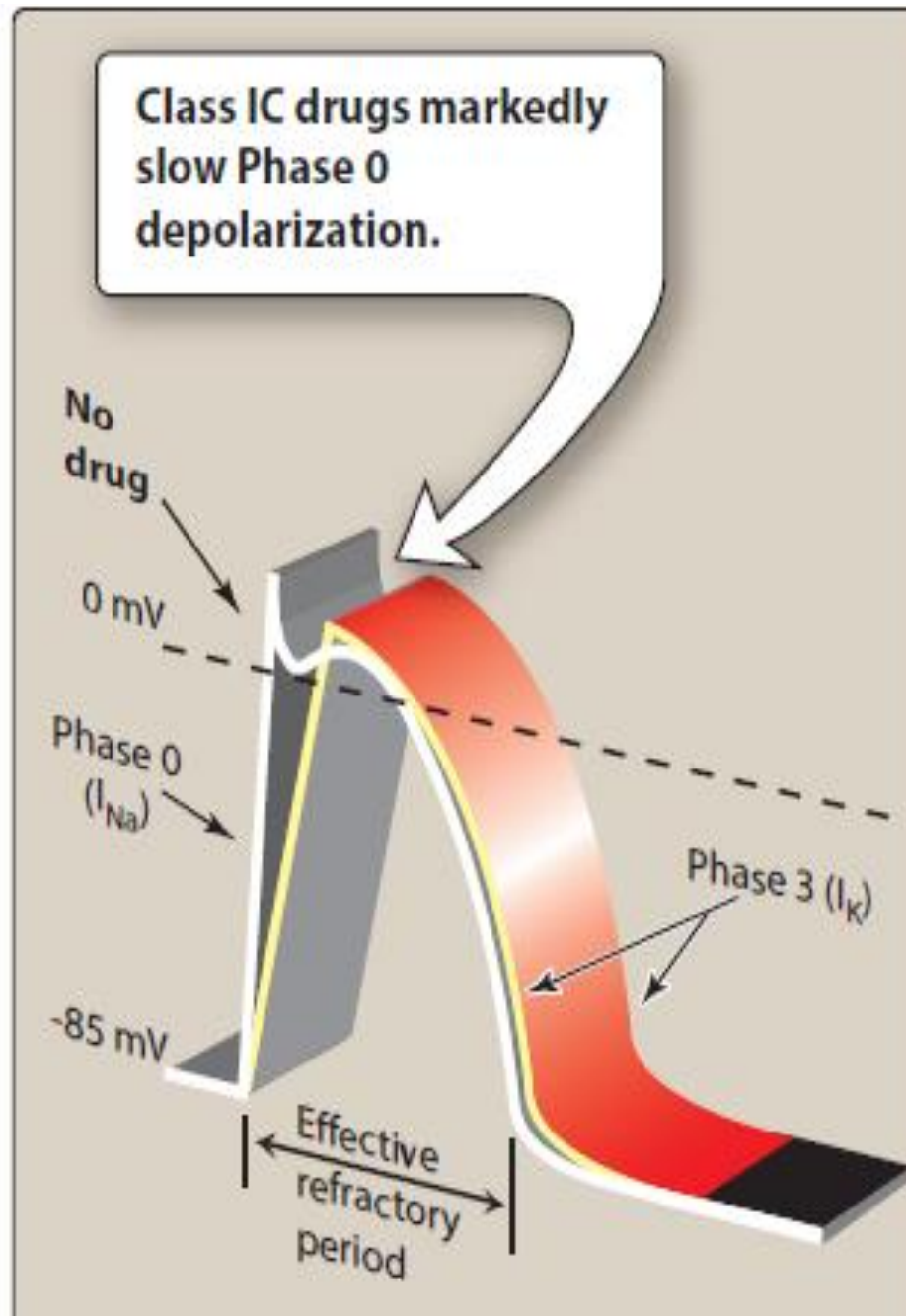


Class IC

- Block fast Na⁺ channels, especially His-purkinje tissue
- No effect on APD and ERP
- No ANS effect

Flecainide

- Limited use because of proarrhythmogenic effects, leading to increased sudden death in post-MI and when used prophylactically in VT



Class II: Beta blockers

- Prevent beta receptor activation, which would normally increase cAMP
- Decreases SA and AV nodal activity
 - Decreases slope of phase 4 (diastolic currents) of AP in pacemakers

Propranolol (nonselective) and cardioselective drugs (acebutolol and esmolol)

Uses:

- prophylaxis in post-MI and in supraventricular tachyarrhythmias (SVTs)
- Esmolol (IV) is used in acute SVTs

Class III: K⁺ channel blockers

- Decrease K⁺ efflux – slowing phase 3 (repolarization) of AP
- Increase APD and ERP, especially in Purkinje and ventricular fibers

Amiodarone

- Mimics class I, II, III, IV
- Increase APD and ERP in all cardiac tissues
- Uses: any arrhythmias
- $t_{1/2} > 80$ days
- Binds extensively to tissues (Large V_d)
- SE: pulmonary fibrosis, Blue pigmentation of the skin, phototoxicity, corneal deposits, Hepatic necrosis, Thyroid dysfunction

Sotalol

- Decreases K⁺ efflux, slowing phase 3
- Beta-1 blockage, leading to decreased HR and AV conduction
- Use: life-threatening ventricular arrhythmia

Class IV: Ca²⁺ channel blockers

- Block slow cardiac Ca²⁺ channels
- Inhibit phase 0 and phase 4, leading to decreased SA and AV nodal activity

Verapamil and diltiazem

- Prototype Ca²⁺ channel blockers
- Uses: supraventricular tachycardias
- SE: constipation (verapamil), dizziness, flushing, hypotension, AV block
- DI: additive AV block with Beta blockers, digoxin;
 - Verapamil displaces digoxin from tissue-binding sites

Unclassified

Adenosine

- Activates Gi-coupled adenosine receptors, causes decrease in cAMP
 - Decreases SA and AV nodal activity
- Adenosine is antagonized by theophylline
- Uses:
 - DOC for paroxysmal supraventricular tachycardias and AV nodal arrhythmias
- Administered IV: $t_{1/2} < 10$ seconds
- SE: flushing, sedation, dyspnea

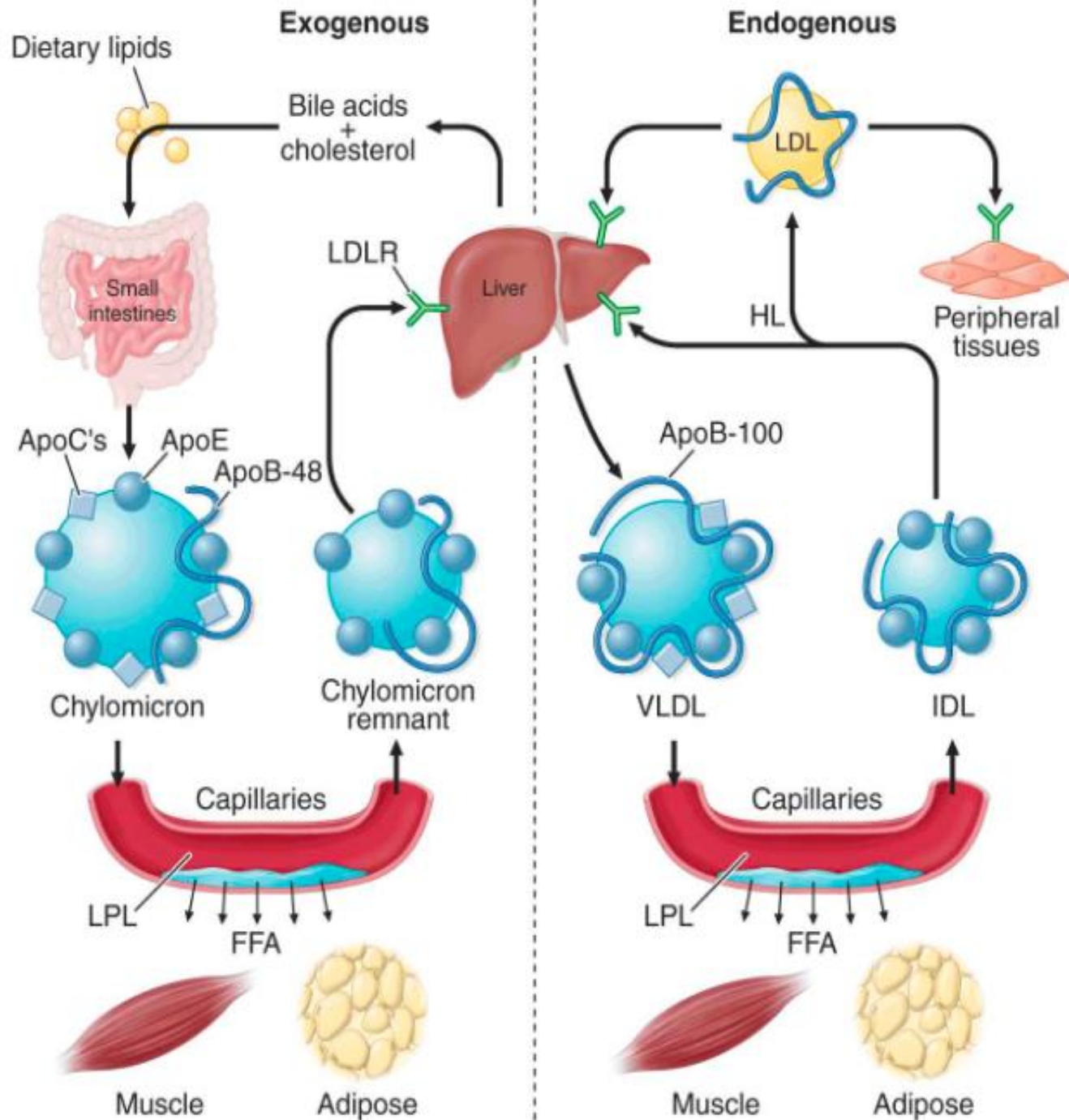
Magnesium

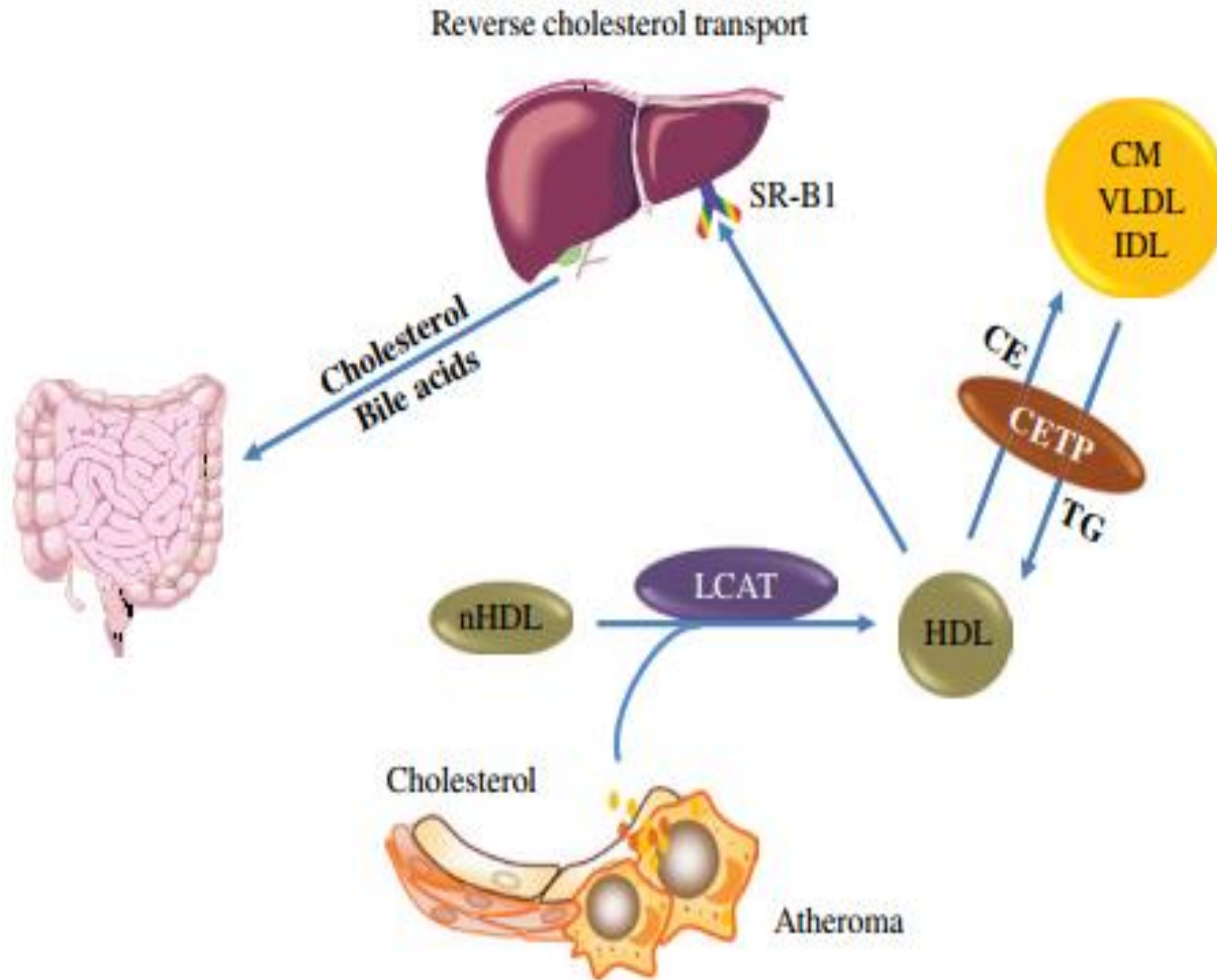
- Use: torsades
- Drugs causing torsades include;
 - K⁺-channel blockers (Class IA and class III)
 - Antipsychotics (thioridazine)
 - Tricyclic antidepressants

Drugs for Dyslipidemias

Dyslipidemias

- Dyslipidemias, including
 - hyperlipidemia (e.g., hypercholesterolemia) and low levels of high-density lipoprotein cholesterol (HDL-C)
- Constitute a chief cause of atherosclerosis and atherosclerosis-associated conditions, such as
 - coronary heart disease, ischemic cerebrovascular disease, and peripheral vascular disease.





LCAT = lecithin-cholesterol acyltransferase

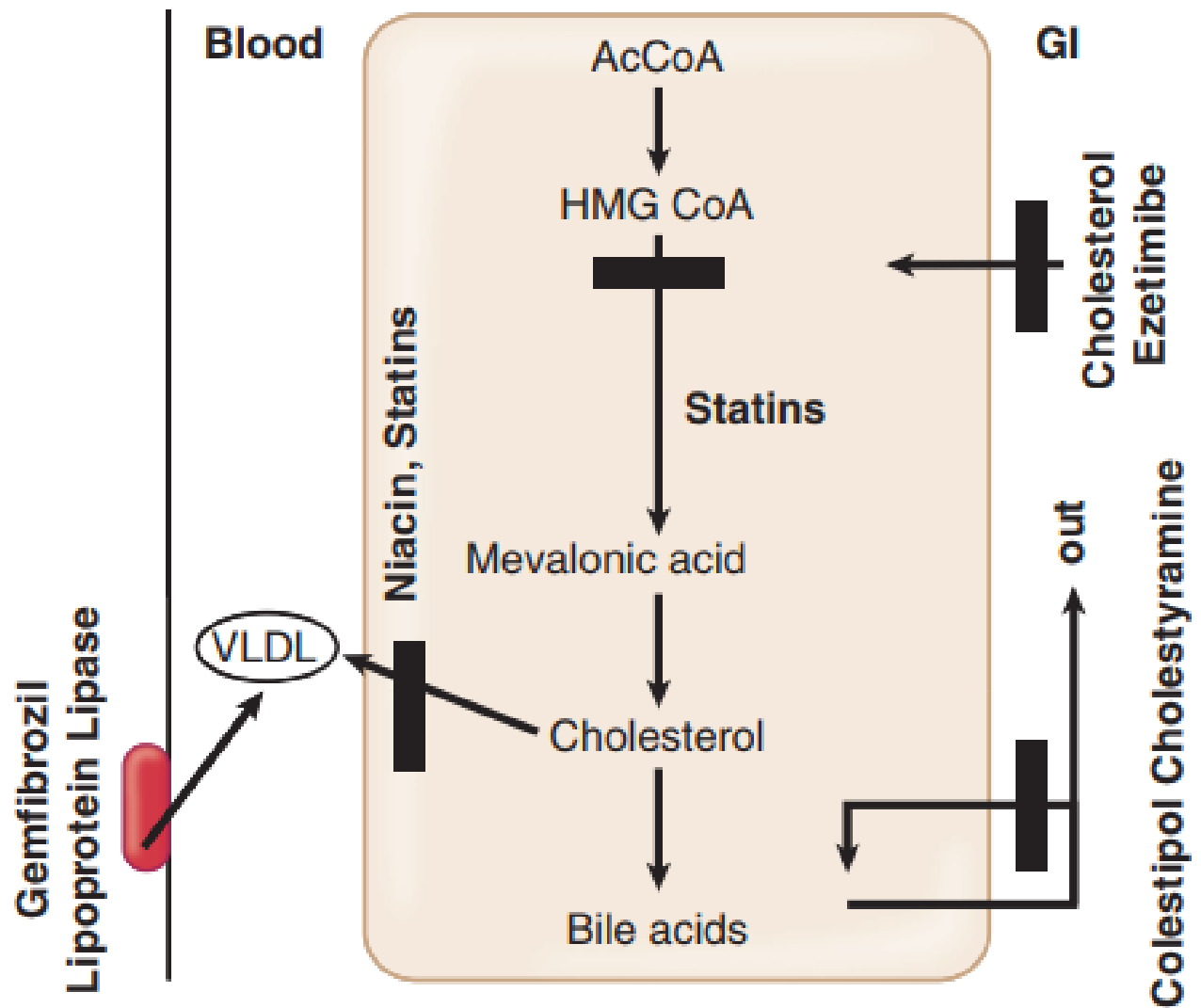
Classification of plasma lipid levels (mg/dl)

Non-HDL-C	
<130	Desirable
130–159	Above desirable
160–189	Borderline high
190–219	High
≥220	Very high
HDL-C	
<40	Low (consider < 50 mg/dL as low for women)
>60	High (desirable because of negative risk)
LDL-C	
<70	Optimal for very high risk ^a
<100	Desirable
100–129	Above desirable
130–159	Borderline high
160–189	High
≥190	Very high
Triglycerides	
<150	Normal
150–199	Borderline high
200–499	High
≥500	Very high

Drugs for dyslipidemia

- Classes of drugs that modify cholesterol levels include:
 - Inhibitors of HMG-CoA reductase (statins)
 - Bile acid-binding resins
 - Fibric acid derivatives (fibrates)
 - Inhibitor of cholesterol absorption (ezetimibe)
 - Nicotinic acid (niacin)
 - PCSK9 inhibitors (alirocumab, evolocumab, and inclisiran)
 - Omega-3 fatty acid ethyl esters (fish oil)
 - MTP inhibitor (lomitapide)
 - Inhibitor of apolipoprotein B-100 synthesis (mipomersen)

Site of Action of anti-dyslipedemic drugs:



HMG-CoA REDUCTASE INHIBITORS

- Drugs: **simvastatin, atorvastatin, rosuvastatin**, and other “**–statins**”
 - At their highest therapeutic doses, atorvastatin and rosuvastatin are considered “high-intensity” statins and can lower LDL-C by $\geq 50\%$
 - Lower doses of statins are classified as “low” or “moderate” intensity
- Mechanisms:
 - HMG-CoA reductase inhibition, results in:
 - $\downarrow$ liver cholesterol
 - $\uparrow$ LDL-receptor expression
 - $\downarrow$ plasma LDL
 - $\downarrow$ VLDL synthesis results in: $\downarrow$ triglyceridemia

- Side effects:
 - Myalgia, myopathy (check creatine kinase)
 - Rhabdomyolysis
 - Hepatotoxicity (check liver function tests)
- Drug interaction:
 - Gemfibrozil (↑ rhabdomyolysis)
 - Cytochrome P450 inhibitors enhance toxicity of statins
- Nonstatin drugs have not been shown to improve cardiovascular outcomes when added to statin therapy.
 - These drugs are most often used in patients who cannot tolerate a statin.

TABLE 33-7 ■ INTENSITY OF STATINS BY APPROXIMATE REDUCTIONS IN LDL-C WITH DAILY DOSING

HIGH-INTENSITY STATINS	MODERATE-INTENSITY STATINS	LOW-INTENSITY STATINS
LOWER LDL-C BY APPROXIMATELY 50% OR MORE	LOWER LDL-C BY APPROXIMATELY 30% TO LESS THAN 50%	LOWER LDL-C, ON AVERAGE, BY LESS THAN 30%
Atorvastatin 40–80 mg Rosuvastatin 20–40 mg	Atorvastatin 10–20 mg Fluvastatin 40 mg twice daily Fluvastatin XL 80 mg Lovastatin 40 mg Pitavastatin 2–4 mg Pravastatin 40–80 mg Rosuvastatin 5–10 mg Simvastatin 20–40 mg	Fluvastatin 20–40 mg Lovastatin 20 mg Pitavastatin 1 mg Pravastatin 10–20 mg Simvastatin 10 mg

BILE ACID SEQUESTRANTS

- Drugs: **cholestyramine** and **colestipol**
- Mechanism: complexation of bile salts in the gut, results in:
 - ↓ enterohepatic recirculation of bile salts
 - ↑ synthesis of new bile salts by the liver
 - ↓ liver cholesterol
 - ↑ LDL-receptor expression
 - ↓ blood LDL
- Side effects:
 - ↑ VLDL and triglycerides
 - Gastrointestinal disturbances
 - Malabsorption of lipid-soluble vitamins
 - Hyperglycemia
- Drug interactions with orally administered drugs (warfarin, thiazides, digoxin, etc.)
- Contraindication: hypertriglyceridemia

NICOTINIC ACID (NIACIN, VITAMIN B3)

- Mechanism: inhibition of VLDL synthesis, results in:
 - ↓ plasma VLDL
 - ↓ plasma LDL
 - ↑ plasma HDL
- Side effects:
 - Flushing, pruritus, burning pain (use aspirin)
 - Hepatotoxicity
 - Hyperglycemia

EZETIMIBE

- Mechanism: prevents intestinal absorption of cholesterol, results in ↓ LDL
- Side effect: gastrointestinal distress

GEMFIBROZIL, FENOFIBRATE (FIBRATES)

- Mechanism: bind to the **PPAR α** and increase expression of **lipoprotein lipases**, results in:
 - ↓ VLDL and IDL
 - Modest ↓ LDL
 - ↑ HDL in most patients
 - In some patients with combined hyperlipidemias, can ↑ LDL
 - Used in hypertriglyceridemia
- Side effects:
 - Gallstones
 - Myositis

OMEGA-3 FATTY ACID ETHYL ESTERS

- **Eicosapentaenoic acid** and **Docosahexaenoic acid** ethyl esters
- reduce VLDL triglycerides
- used as an adjunct to diet for treatment of adult patients with severe hypertriglyceridemia.
- The recommended daily oral dose for patients with severe hypertriglyceridemia is
 - 3-4 g/d administered with food.
- Adverse effects: arthralgia, nausea, fishy burps, dyspepsia, and increased LDL
- Because omega-3 fatty acids may prolong bleeding time,
 - patients taking anticoagulants should be monitored

PCSK9 INHIBITORS

- **Alirocumab, evolocumab, and inclisiran**
- MOA: Proprotein convertase subtilisin/kexin type 9 (PCSK9) is a protease that binds to the LDL receptor on the surface of hepatocytes and
 - enhances lysosomal degradation of the LDL receptor, resulting in higher plasma LDL concentrations.
- PCSK9 inhibitors enhance LDL receptors to be available on the surface of liver cells.
- PCSK9 inhibitors reduce LDL-C in a dose-dependent manner
 - by as much as 70% when used as monotherapy or by as much as 60% in patients already on statin therapy.

INHIBITOR OF MICROSOMAL TRIGLYCERIDE TRANSFER PROTEIN

- **Lomitapide**; inhibiting MTP, which is essential for the formation of VLDLs
- reduces LDL by up to 50% and
 - should be used in combination with maximally tolerated statin therapy.

INHIBITOR OF APOLIPOPROTEIN B-100 SYNTHESIS

- **Mipomersen**; binds to the mRNA of apo B-100 in a sequence-specific manner, which results in
 - degradation or disruption of the apo B-100 mRNA, thereby reducing expression of apo B-100 protein.

END